

Elementary Particle Physics

- A quest to find what are matters made of?
- Probing in smaller scale (higher energy) Atoms $\rightarrow$ nucleus + electrons $\rightarrow$ protons, neutrons
- Not only lighter particles were discovered, heavy particles were discovered as well
- They point out to a "structure" controlling the physics at small scales
(Quantum field theory)
- The structure leads to the unification scheme of physics
(electric + magnetic, electroweak, Grand Unified theory, string/M?)
- In this course we will explore physics of elementary particles.
 - How do we know they exist?
 - What are their properties? / interactions?
 - What is the underlying structure? and what is the physics describing it?

} ~90% of the course

Discoveries of particles

Electrons ~1897



• J.J. Thompson, cathode ray experiment

$$F = eE, F = ma, a = \frac{e}{m} E$$

- Measuring how far an electron is deflected $\Rightarrow a$, plot a vs E we find $\frac{e}{m}$
- $\frac{e}{m} = 1.8 \times 10^{11} \text{ C/kg}$
- Millikan's droplet experiment
 - Using Stoke's law on oil drop $\Rightarrow$ Weight of droplets
 - Turn $F = qE$, $E = \frac{V}{d} \Rightarrow$ And that q 's are always an integer of
 $e = 1.6 \times 10^{-19} \text{ C}$

Proton

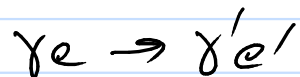
- Rutherford scattering 1910
- Atoms consist of electrons & positive charge nuclei
- Positive charge is concentrated in volume of the size which is much smaller than the size of atom

Neutron

- Chadwick 1932
- Using an unknown radiation from $\alpha + Be \rightarrow n + X$ to scatter of the ionized chamber filled with protons
$$n + p \rightarrow n + p'$$
- Collecting the angular distribution of p' , he found that
$$m_n \approx m_p$$

Photon

Compton effect - scattering of x-rays on electrons (in crystals)

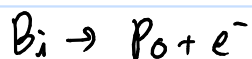


x-rays change frequency $f \rightarrow f' \Rightarrow$ can be explained

by assuming that photon is a particle with the energy $E = hf$ and momentum $p = \frac{hf}{c}$

Neutrinos

- ν was first introduced by Pauli (1930) $\Rightarrow$ explain properties of β -decay



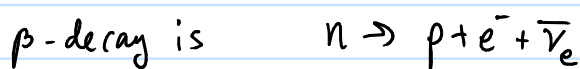
- We expect discrete energy spectrum (neglect the recoil energy)

$$E_e = Q = M(\text{Bi}) - M(\text{Po})$$

Instead Chadwick found continuous E_e spectrum

- Spin is not conserved! $\text{Bi} \rightarrow$ Bose statistics, angular momentum $(\text{Po} + e^-) = \frac{1}{2}$
- A proposal of a new particle with following properties
 - 1). sufficiently light ($m < 0.1 \text{ MeV}$)
 - 2). electrically neutral
 - 3). spin $\frac{1}{2}$

Pauli hypothesis has been included into Fermi theory of weak interaction



with Fermi interaction
$$H = \frac{G_F}{\sqrt{2}} \underbrace{\bar{p}(x) \Gamma_{\text{had}}^\mu n(x)}_{\text{Hadronic current}} \underbrace{\bar{e}(x) \Gamma_{\text{lep}}^\mu \nu(x)}_{\text{Leptonic current}} + \text{h.c.}$$

- These currents have V-A structure (vector - axial vector)

$$\Gamma^\mu = \gamma^\mu (1 - \gamma_5)$$

[We will come back to this point after we learnt more about Dirac spinor]

- G_F has mass dimension -2, $[G_F] = \frac{1}{m^2}$

$$G_F \sim 10^{-5} / m_{\text{proton}}^2$$

Discovery of $\bar{\nu}_e$

- Cowan & Reines (1956)

- nuclear reactor as a source of $\bar{\nu}_e$.
- The process is $\bar{\nu}_e + p \rightarrow n + e^+$
- the tank filled with the water solution of CdCl_2
- (anti)neutrinos come in the tank, interacts with protons in the water
- The positron annihilates with electrons producing photon which is detected by photomultipliers
- The neutron is bouncing with protons until it gets absorbed by Cd
$$n + \text{Cd} \rightarrow \text{Cd} + \gamma\gamma\gamma \dots$$

producing secondary flash of light in the detector.

Positron

- Predicted by Dirac (1928)
- Discovered by Anderson (1932) using Wilson cloud chamber

Wilson Chamber:

- Phase transition gas $\rightarrow$ liquid in the supersaturated gas
 $\Rightarrow$ a small disturbance can lead to the phase transition (condensation)

From photographs, we can see the tracks indicating:

- Curvature (under the presence of magnetic field)
= momentum / charge

$$e\cancel{v}B = \frac{mv^2}{R}$$
$$R \sim \frac{p}{e}$$

- Thickness of the track

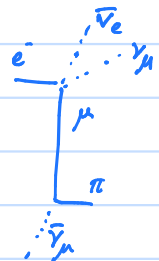
- Comparison of tracks before & after passing the plate (lead)
- Positron was discovered from (secondary) cosmic ray
- The tracks show that $\left\{ \begin{array}{l} p \sim 20 \text{ MeV} \\ e = +1 \end{array} \right\}$ but the energy loss in Pb is low
 $\Rightarrow$ cannot be proton. $\Rightarrow$ new particle

Muon

- Anderson 1936
- A charged track with thin track, also easily penetrates Pb
- $p \sim 200-300 \text{ MeV}$
- cannot be proton (non-relativistic $\Rightarrow$ thick track)
- cannot be electron (relativistic $\Rightarrow$ large energy loss in Pb plate)
- The track can be explained if the particle has mass $m \sim \frac{1}{10} m_p$
- also unstable $\tau_\mu \sim 2 \times 10^{-6} \text{ sec}$ ($m_\mu \sim 105 \text{ MeV}$)

Pion

- was discovered in nuclear emulsion chamber (similar to cloud chamber)
- Pion loses energy and decays $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$
- then μ stops and decays $\mu \rightarrow \nu_\mu + \bar{\nu}_e + e^-$
- $m_\pi \approx 140 \text{ MeV}$, $\tau_\pi \sim 2 \times 10^{-8} \text{ sec}$
- Also π^0 has been discovered through $\pi \rightarrow \gamma\gamma$
- It turns out that the Hamiltonian for pion decay is similar to β^- decay



$$H = \frac{G_F}{\sqrt{2}} \bar{e} \gamma^\mu (1 - \gamma_5) \nu_e \cdot \bar{\nu}_\mu \gamma_\mu (1 - \gamma_5) \mu$$

- Pions are force mediators of nuclear force inside nuclei
- The strong interaction has a finite range

$$\phi \propto \frac{e^{-r/\lambda}}{r}$$

- $m \sim 140 \text{ MeV} \Rightarrow \lambda \sim \frac{1}{m} = 1.4 \text{ fm}$ (effective range of st. int.)

Strange particles

- K-mesons were discovered in 1947 (Rochester, Butler)
- Cosmic ray in cloud chamber;
curvature & thickness $\Rightarrow m_K \sim 500 \text{ MeV}$, $\tau_K \sim 1.2 \times 10^{-8} \text{ sec}$
- $K \rightarrow \mu \nu_\mu$
- They have "strange" properties: produced in strong interaction, but decay only through weak interaction
- $\# K \text{ tracks} \sim \# \text{ pion tracks} \Rightarrow$ produced in strong interaction
- But there is no fast decay like $K \rightarrow \pi\pi$ (strong int) (It would have been 10^{-23} sec)
- We need to introduce the new conservation laws / quantum number
- The strangeness $S(K^+) = S(K^0) = +1$, $S(K^-) = S(\bar{K}^0) = -1$
is conserved in strong interaction but violated in weak interaction
- $\Rightarrow$ There are also baryon & lepton numbers
- Kaons also led to charm quark indirect discovery (GIM mechanism)

Quarks

- A large number of strongly interacting particles have been discovered. (hadrons)
- The baryon number; $B=0 \Rightarrow$ mesons, π, K, ρ / $B=\pm 1 \Rightarrow$ baryons, p, n, Λ, Σ

• Isospin

- notice that $m_p \approx m_n$ and strong force are almost identical for p and n
- Heisenberg introduced isospin concept that p & n form the doublet $\begin{pmatrix} p \\ n \end{pmatrix}$
- p and n are two components of a single particle (nucleon). They correspond to different third projections of the isospin operator $I_3(p) = \frac{1}{2}$, $I_3(n) = -\frac{1}{2}$

- Isospin & electric charge

- one can try $Q = I_3 + \frac{B}{2}$

which works on $\begin{pmatrix} p \\ n \end{pmatrix}$, $\begin{pmatrix} \pi^+ \\ \pi^0 \\ \pi^- \end{pmatrix}$

- However for Kaons $\begin{pmatrix} K^+ \\ K^0 \end{pmatrix}$, $\begin{pmatrix} \bar{K}^0 \\ K^- \end{pmatrix}$, we need to consider strangeness

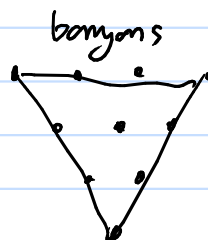
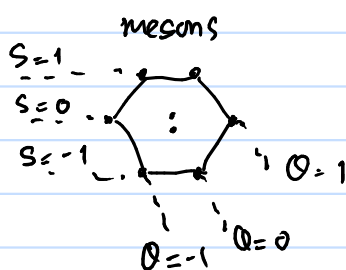
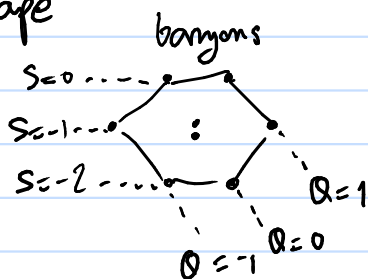
$$Q = I_3 + \frac{B+S}{2}$$

- $B+S$ can be defined as the hypercharge $Y = B+S$

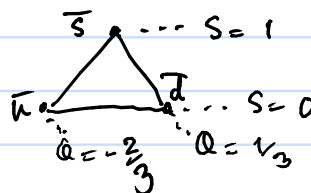
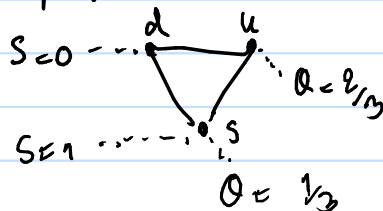
$$Q = I_3 + \frac{Y}{2} \quad (\text{Gellmann-Nishijima formula})$$

The eightfold way

- Gellmann notice that baryons and mesons can be arranged into a hexagonal shape

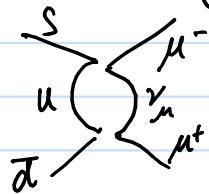


- Gellman in 1964 proposed that all hadrons are composed of quarks

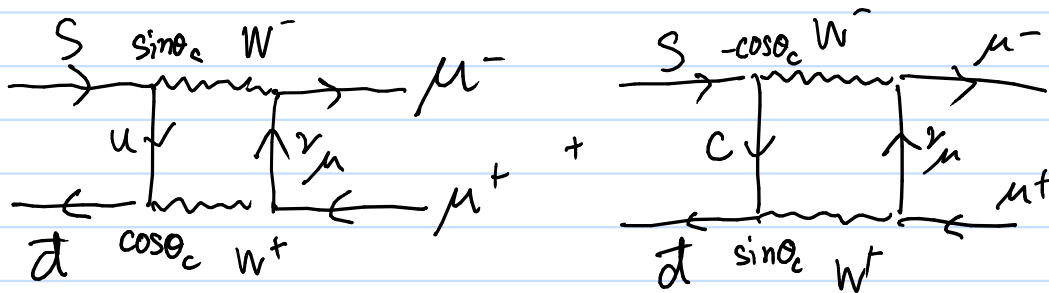


Charm quark and GIM mechanism

- At this point in history there are $\{u, d, s, e, \nu_e, \mu, \nu_\mu\}$
- To complete the doublet structure (as a weak current), we need a new quark
 $\begin{pmatrix} u \\ d \end{pmatrix}, \begin{pmatrix} c \\ s \end{pmatrix}, \begin{pmatrix} \nu_e \\ e \end{pmatrix}, \begin{pmatrix} \nu_\mu \\ \mu \end{pmatrix}$
- Introduce charm quark $\Rightarrow$ explain an absence (strong suppression) of the flavor changing neutral currents (FCNC). The kaon decay $K^0 \rightarrow \mu^+ \mu^-$ is supposed to be much higher than experimental result.



This can be explained by adding a new diagram



θ_c is Cabibbo mixing angle. The experiment result can be matched if $m_u \neq m_c < 1-2 \text{ GeV}$

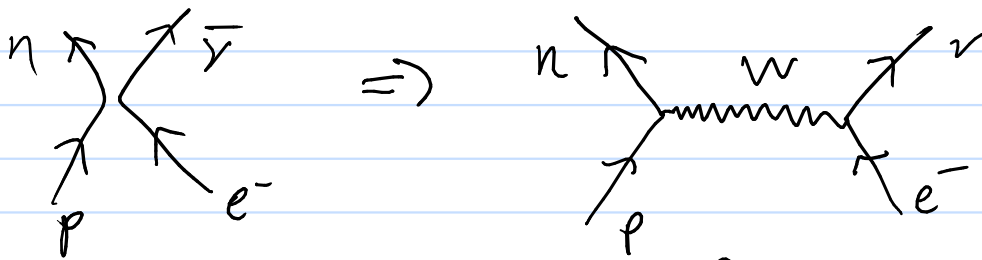
Discoveries in collider

- charmonium (J/ψ), tau lepton, b-quark mesons (Υ)

Discovery of the intermediate bosons

- W bosons were introduced in an analogy of weak int. with electrodynamics
- W bosons should be massive (looks like a point-like interaction at low energies)
 - charged (to couple with charged currents)
 - parity violating (having V-A structure $\Rightarrow$ left handed only)

$$H_{\text{eff}} = \frac{G_F}{\sqrt{2}} J^\mu \cdot J_\mu \Rightarrow H = \frac{g}{2\sqrt{2}} J_\mu \cdot W^\mu + \text{h.c.}$$



$$\frac{G_F}{\sqrt{2}} = \left(\frac{g}{2\sqrt{2}}\right)^2 \frac{1}{M_W^2} = \frac{g^2}{8M_W^2}$$

- Why V-A is left handed (more theoretical approach will come later)

$$f_L \equiv \frac{(1-\gamma_5)}{2} f, \quad f_R \equiv \frac{(1+\gamma_5)}{2} f$$

$$\bar{\nu} \gamma^\mu (1-\gamma_5) l \sim \bar{\nu}_L \gamma^\mu l_L$$

- The model predicts $g = e/\sin\theta_w$. Solving for m_w gives

$$m_W^2 = \frac{\alpha \cdot \pi}{\sqrt{2} G_F \sin\theta_w}; \quad \alpha = \frac{e^2}{4\pi}$$

- Z^0 appeared as an outcome of weak + EM

- Interactions are fixed by the model:

- Z^0 has interaction with left-handed and right-handed

- Parity is violated $\Rightarrow$ interactions for f_L, f_R are different

$$m_Z = \frac{m_W}{\cos\theta_w}$$

Discovery of W & Z

$$p + \bar{p} \rightarrow W^\pm + X, \quad \bar{p} + p \rightarrow Z^0 + X$$

$$W \rightarrow l + \nu_l, \quad Z \rightarrow l^+ l^-$$

- They are detected by CERN sps collider

Gluons

- Δ^{++} has 3 u quarks. quarks are fermions $\Rightarrow$ Pauli principle?
- A new quantum number was proposed $\Rightarrow$ color = {R, G, B}
- The calculation of $e^+ + e^- \rightarrow q + \bar{q} + \text{hadrons}$ gives a factor of $(3)^2$ short of experiment $\Rightarrow$ color $\Rightarrow$ new degree of freedom
- The discovery of the mediators of color force (gluons) comes from the 3-jets events in DESY

The Standard Model (SM) of elementary particle physics (The Glashow-Weinberg-Salam model)

- It is the theory that describes the strong, weak and EM interactions of elementary particles, also predicts the unification of weak + EM
- The first generation of particles constitutes ordinary matters

		Q	M (MeV)
Quarks	u	2/3	$\sim 1-10$
	d	-1/3	$\sim 1-10$
Leptons	ν_e	0	$\leq 10^{-6}$
	e	-1	0.5

- They are spin $1/2$ fermions
- Quarks also have color charge (they feel strong force while leptons are neutral under it)
- There are 2 additional sets of replica called 2nd, 3rd family

		Q	M (MeV)
Quarks	c	2/3	1500
	s	-1/3	100
Leptons	ν_μ	0	$\leq 10^{-6}$
	μ	-1	100

		Q	M (MeV)
Quarks	t	2/3	174300
	b	-1/3	4500
Leptons	ν_e	0	$\leq 10^{-6}$
	e	-1	1780

• There are 3 interactions

1) Electromagnetic

- is very accurately described by a quantum field theory model based on a $U(1)$ gauge group
- } Quantum Electrodynamics (QED)

2) Strong interaction

(aka color interaction)

- It is felt by quarks only
 - Strong forces produce bound states of 3 quarks / quark-antiquark pairs / penta quark?
- } Asymptotic freedom
- Strong interactions can be modeled by a Yang-Mills gauge theory based on a gauge group $SU(3)_c$ [Quantum Chromodynamic or QCD]

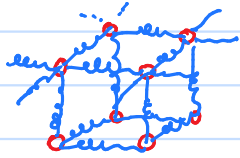
3) Weak interaction

- responsible for β decay
- based on $SU(2)$ gauge group
- massive $W^\pm, Z$ bosons signal symmetry breaking which then leads to the Higgs mechanism

The essence of the SM:

- 1). The elementary particles are quarks and leptons
- 2). The mathematical framework for forces are gauge theories
- 3). The vacuum is in a superconducting phase (Cooper pair ~ Higgs)

Fields

- To understand the mathematical structure of particle physics, it is essential to use the concept of field
- The field is an entity that is defined over all space and time
Ex. $\phi(x,t)$, $\psi(x,t)$, $E(x,t)$, $\vec{B}(x,t)$ (Scalar, vector fields, tensor field)
- In this language particle = energy quanta of a field
- The analogy can be made by a system of atoms connected by springs
 $\Rightarrow$ Atoms can vibrate and transmit vibration to neighboring lattices
- The quantum of the lattice vibration = phonon
- Take the lattice is infinitely large & the lattice spacing is so small
 $\Rightarrow$ phonon is a real particle

Natural units

- Natural units = a system of units which help thinking about physics easier
- We take $c = 1$, $\hbar = 1$ (also $k_B = 1$ but not relevant to us now)
 $G = 1$
- What does this implies? $\Rightarrow$ Everything boil down to 1 unit.

- Since $c = 3 \times 10^8 \text{ m/s} = 1$

$$\Rightarrow 1 \text{ s} \approx 3 \times 10^8 \text{ m}$$

- $\hbar = 1.05 \times 10^{-34} \text{ J s} = 1$

$$\Rightarrow 1 \text{ J} \approx 10^{34} \text{ s}^{-1}$$

$$\text{Since } 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J} = 1.6 \times 10^{15} \text{ s}^{-1}$$

- Now we can relate any units in this framework

$$1 \text{ m} = \frac{1}{3} \times 10^{-8} \text{ s} = \frac{1}{3} \times 10^{-8} \times 1.6 \times 10^{15} \text{ eV}^{-1} \sim 0.5 \times 10^7 \text{ eV}^{-1}$$

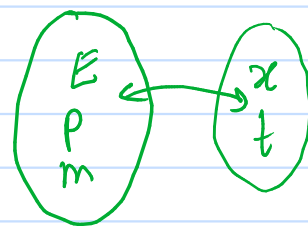
$$\text{or } 1 \text{ m} = 1 \times \frac{1}{c} \frac{\text{m} \cdot \text{s}}{\text{m}} = 1 \times \frac{1}{c} \times \frac{1}{\hbar} \frac{\text{m} \cdot \text{s}}{\text{m} \cdot \text{J s}} = \frac{1}{c \hbar} \times 1.6 \times 10^{-19} \frac{\text{m} \cdot \text{s}}{\text{m}} \times \frac{1}{\text{J s}} \times \frac{\text{J}}{\text{eV}}$$

$$= \frac{1.6 \times 10^{-19}}{3 \times 10^8 \times 1 \times 10^{-34}} \text{ eV}^{-1} = 0.5 \times 10^7 \text{ eV}^{-1}$$

- Moreover as the extension of eV $\left\{ \begin{array}{l} 1 \text{ keV} = 10^3 \text{ eV}, 1 \text{ MeV} = 10^6 \text{ eV} \\ 1 \text{ GeV} = 10^9 \text{ eV}, 1 \text{ TeV} = 10^{12} \text{ eV} \end{array} \right\}$

★ The natural units allow us to convert between energy $[E]$, $1/\text{time} [T]^{-1}$ and $1/\text{distance} [L]^{-1}$

$\Rightarrow$ This is very useful for dimensional analysis



Example In Rutherford's scattering, given that the velocity of α particles

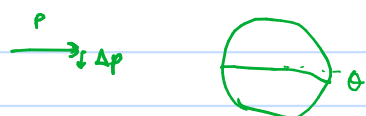
are $v \sim 0.03$ (10^4 km/s), find the typical scattering angle if positive charges

are evenly distributed in the size of $r = 10^{-8} \text{ cm}$

- The typical momentum transfer $\sim \Delta p \sim 1/r \sim 10^6 \text{ GeV}$

- The momentum of $\alpha = mv = 4 \times 0.03 = 0.12 \text{ GeV}$

$$\theta \sim \frac{\Delta p}{p} \sim 10^{-5} \text{ (very small deflecting angle)}$$



Now try out $r \sim 10^{-13} \text{ cm}$

What is an angle?

Homework

- 1). Find $\frac{1}{m_{e^-}}$ in metre (This is Compton wavelength of e^-)
- 2). Find $\frac{1}{m_p}$ in metre (This is the characteristic size of a nucleus)
- 3). Find the probing scale (in metre) of the LHC at 14 TeV
- 4). Express Newtons constant G_N in eV^n and s^m
- 5). Compare the relative strength of F_{em} & $F_{gravity}$ between e^+ & e^-
- 6). Approximate the effective ranges of the weak force, the strong force between nucleus and the Higgs force.
- 7). Estimate the number of protons in the Atlantic ocean.
(You are allowed to use google for relevant information)

Particle Physics Homework 1

1) Beirut Explosion

On 4th August 2020, an explosion occurred at Beirut, the capital city of Lebanon. We can use the dimensional analysis technique developed by Geoffrey Taylor (1945) to estimate the total energy output of the bomb.

The analysis can be done by proposing that there are only 4 parameters in the physics of explosion:

E = total energy output, t = time

R = blast radius, ρ = air density.

We can work out the equation as

$$R = E^{\alpha} \rho^{\beta} t^{\gamma}$$

1.1) Work out the power α , β , and γ as a number using dimensional analysis (no need for natural unit)

1.2) The Beirut's explosion was captured by video and one of the frame has been analysed such that

At $t = 0.233$ s, the blast radius is 179.97 m.

The air density at that time is 1.23 kg/m^3 and

1 million kilogram of TNT (kt) = $4.18 \times 10^{12} \text{ J}$.

Find the energy output in kt.

2). Natural Unit

- 2.1). Find $\frac{1}{m_e}$ in metre (This is Compton wavelength of e^-)
- 2.2). Find $\frac{1}{m_p}$ in metre (This is the characteristic size of a nucleus)
- 2.3). Find the probing scale (in metre) of the LHC at 14 TeV
- 2.4). Express Newtons constant G_N in eV^n and s^m
- 2.5). Compare the relative strength of F_{em} & $F_{gravity}$ between e^+ & e^-
- 2.6). Approximate the effective ranges of the weak force, the strong force between nucleus and the Higgs force.
- 2.7). Estimate the number of protons in the Atlantic ocean.
(You are allowed to use google for relevant informations)

3). Vector Calculus

Identities in Vector Calculus are a lot easier to prove in the index form. Let's learn how to prove them in this exercise.

The notation of vector in the index form is given by:

$$\vec{A} \cdot \vec{B} = A_i B_i \quad (\text{inner product})$$

$$(\vec{A} \otimes \vec{B})_{ij} = A_i B_j \quad (\text{tensor product})$$

$$(\vec{A} \vec{x})_i = A_{ij} x_j \quad (\text{matrix multiplication})$$

$$\text{tr}(A) = A_{ii} \quad (\text{trace})$$

$$(A^T)_{ij} = A_{ji} \quad (\text{transpose})$$

Another useful symbol is the Levi-Civita symbol, ϵ_{ijk} , defined as

i) $\epsilon_{ijk} = 0$ if any two indices are equal ($\epsilon_{112} = \epsilon_{131} = \epsilon_{332} = 0$)

ii) $\epsilon_{ijk} = -\epsilon_{jik} = -\epsilon_{kji} = -\epsilon_{ikj}$

(It picks up a minus sign for any odd number of permutation

e.g. $\epsilon_{123} \xrightarrow{1 \text{ time}} -\epsilon_{213} \xrightarrow{2 \text{ times}} \epsilon_{231}$

iii) $\epsilon_{123} = 1$

This symbol is useful for writing the cross product

Consider 2 vectors: $\vec{A} = A_i \hat{e}_i$, $\vec{B} = B_i \hat{e}_i$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \hat{e}_1(A_2 B_3 - A_3 B_2) + \hat{e}_2(A_3 B_1 - A_1 B_3) + \hat{e}_3(A_1 B_2 - A_2 B_1)$$

$\uparrow \epsilon_{123} \quad \uparrow \epsilon_{132} \quad \uparrow \epsilon_{231} \quad \uparrow \epsilon_{213} \quad \uparrow \epsilon_{312} \quad \uparrow \epsilon_{321}$

$$= \epsilon_{ijk} \hat{e}_i A_j B_k \quad \therefore (\vec{A} \times \vec{B})_i = \epsilon_{ijk} A_j B_k$$

The product of 2 Levi-Civita symbols is written as

$$\epsilon_{ijk} \epsilon_{lmn} = \delta_{il} \delta_{jm} \delta_{kn} + \delta_{im} \delta_{jn} \delta_{kl} + \delta_{in} \delta_{jl} \delta_{km} - \delta_{im} \delta_{jl} \delta_{kn} - \delta_{il} \delta_{jn} \delta_{km} - \delta_{in} \delta_{jm} \delta_{kl}$$

Now prove the following identities:

3.1) $\epsilon_{ijk} \epsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}$

3.2) $\epsilon_{ijk} \epsilon_{ijn} = 2 \delta_{kn}$

3.3) $\epsilon_{ijk} \epsilon_{ijk} = 6$

3.4) $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$

$$3.5) \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A}$$

$$3.6) \vec{\nabla} \times (\vec{\nabla} f) = 0$$

$$3.7) \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

$$3.8) \vec{\nabla} \times (\vec{A} \times \vec{B}) = \vec{A}(\vec{\nabla} \cdot \vec{B}) - (\vec{\nabla} \cdot \vec{A})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B}$$

$$3.9) \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{A} \cdot (\vec{\nabla} \times \vec{B}) - \vec{B} \cdot (\vec{\nabla} \times \vec{A})$$

$$3.10) \vec{\nabla}(\vec{A} \cdot \vec{B}) = \vec{A} \times (\vec{\nabla} \times \vec{B}) - (\vec{\nabla} \times \vec{A}) \times \vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} + (\vec{A} \cdot \vec{\nabla})\vec{B}$$

$$3.11) \vec{A} \times (\vec{B} \times \vec{C}) + \vec{C} \times (\vec{A} \times \vec{B}) + \vec{B} \times (\vec{C} \times \vec{A}) = 0$$

$$3.12) \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$3.13) (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{B} \cdot \vec{C})(\vec{A} \cdot \vec{D})$$

$$3.14) (\vec{A} \cdot (\vec{B} \times \vec{C}))\vec{D} = (\vec{A} \cdot \vec{D})(\vec{B} \times \vec{C}) + (\vec{B} \cdot \vec{D})(\vec{C} \times \vec{A}) + (\vec{C} \cdot \vec{D})(\vec{A} \times \vec{B})$$

$$3.15) (\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = (\vec{A} \cdot (\vec{B} \times \vec{D}))\vec{C} - (\vec{A} \cdot (\vec{B} \times \vec{C}))\vec{D}$$

Lorentz invariance (Special Relativity)

(natural unit remember!)

$$v \rightarrow 1$$

• In particle physics, all particles are relativistic $E \approx p \gg m$

• The theory of special relativity $\Rightarrow$ space & time are intertwined

2 principles of SR

1). Invariance of the speed of light: c always remains the same in any inertial frame

2). Relativity principle: physical laws must be invariant in any inertial frame.

Vectors

We write a vector $\vec{r}$ in 2 ways

(you can choose $r_1=x, r_2=y, r_3=z$ and get back to Cartesian coord.)

1) component form $\vec{r} = r_1 \hat{e}_1 + r_2 \hat{e}_2 + r_3 \hat{e}_3 \equiv r_i \hat{e}_i$

where we use "Einstein Summation Convention"

★ This is particularly useful for scalar product (dot product, inner product)

$$\vec{A} \cdot \vec{B} = A_i B_i = A_1 B_1 + A_2 B_2 + \dots + A_n B_n, \quad i=1, \dots, n$$

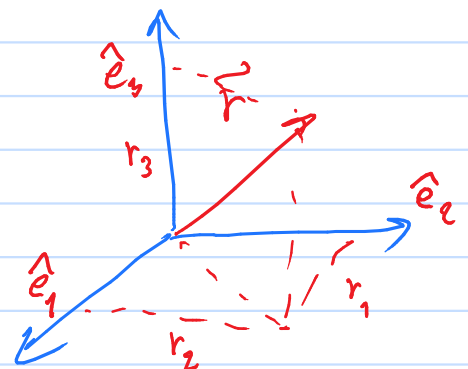
Define Kronecker delta: $\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$ Find $\delta_{ii} = ?$

$$\delta_{ii} = \underbrace{1 + 1 + \dots + 1}_n = n$$

Find $\delta_{ik} \delta_{kj} = ?$

$$\begin{aligned} \delta_{ik} \delta_{kj} &= \delta_{i1} \delta_{1j} + \delta_{i2} \delta_{2j} + \dots + \delta_{in} \delta_{nj} \\ &= \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases} = \delta_{ij} \end{aligned}$$

$$\vec{A} \cdot \vec{B} = A_i \delta_{ij} B_j = A_i B_i = A_1 B_1 + \dots + A_n B_n$$



2). Matrix form $\vec{r} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = r_i \hat{e}_i$

The scalar product $\vec{A} \cdot \vec{B} = (A_1 \ A_2 \ A_3 \ \dots \ A_n) \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{pmatrix} = A_i B_i = A_1 B_1 + \dots + A_n B_n$

$\underbrace{\hspace{10em}}_{A_i} \quad \underbrace{\hspace{1em}}_{B_i}$

Also we can write $\vec{A} \cdot \vec{A} = A_i \delta_{ij} B_j = (A_1 \ A_2 \ \dots \ A_n) \begin{pmatrix} 1 & 0 & & 0 \\ 0 & 1 & & 0 \\ & & \ddots & \\ 0 & & & 1 \end{pmatrix} \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{pmatrix}$

Rotation

Rotations are transformations of coordinates which does not change the length of any vector

The rotation can be written as $\vec{A}' = \overset{\text{matrix}}{\underline{R}} \vec{A} = R_{ij} A_j \hat{e}_i = \begin{pmatrix} R_{11} & R_{12} & \dots \\ R_{21} & & \\ \vdots & & R_{nn} \end{pmatrix} \begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix} = \begin{pmatrix} A'_1 \\ \vdots \\ A'_n \end{pmatrix}$

The definition of rotation implies the properties of $\underline{R}$ or R_{ij}

Since $\vec{A}' \cdot \vec{A}' = \vec{A} \cdot \vec{A}$

$$(A_1 \ \dots \ A_n) \begin{pmatrix} R_{11} & R_{12} & \dots \\ R_{21} & & \\ \vdots & & R_{nn} \end{pmatrix}^T \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \begin{pmatrix} R_{11} & R_{12} & \dots \\ R_{21} & & \\ \vdots & & R_{nn} \end{pmatrix} \begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix} = (A_1 \ \dots \ A_n) \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix}$$

$$\Rightarrow \boxed{\underline{R}^T \underline{1} \underline{R} = \underline{1}} \quad (\underline{R}^T \underline{R} = \underline{1})$$

In component form

↳ "1" is telling us the geometry of the space (Euclidean)

$$A'_j \delta_{jk} A'_k = A_i R_{ij} \delta_{jk} R_{kl} A_l = A_i \delta_{ij} A_j = A_i \delta_{il} A_l$$

$$\Rightarrow \boxed{R_{ij} \delta_{jk} R_{kl} = \delta_{il}} \quad (R_{ij} R_{jl} = \delta_{il})$$

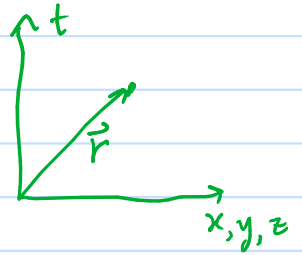
Lorentz transformation

(4-vector)

Now consider the space-time vector $\vec{r} = r^\mu \hat{e}_\mu = (ct, x, y, z)$

(From now on we will use Greek indices for the 4-vector and roman indices for a vector) $[\mu, \nu, \sigma, \rho = 0, 1, 2, 3 / i, j, k = 1, 2, 3]$

natural unit
common notation



This is the spacetime position vector of the event E_1

The scalar product which is consistent with Einstein's postulation is the scalar product in Minkowski space

$$\vec{A} \cdot \vec{B} \equiv (A_0, A_1, A_2, A_3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \end{pmatrix} = A^\mu g_{\mu\nu} B^\nu = A_\mu B^\mu$$

upstairs or downstairs?

- The way we use both positions now is just to separate 4-vectors with a normal one.
- "Contravariant" = A^μ
- "Covariant" = $A_\mu \equiv A^\nu g_{\nu\mu}$
- In this way, the scalar product is composed of $A_\mu A^\mu$

• $g_{\mu\nu}$ is called the metric tensor

• The Lorentz transformation is a set of transformations that does not change the scalar product of the Minkowski

space ($A_\mu A^\mu$ is invariant)

$\Rightarrow$ Assume that $A^{\mu'} = \Lambda^{\mu'}_\nu A^\nu$, we obtain the property of Lorentz transt.

$$A^{\mu'} g_{\mu'\nu'} A^{\nu'} = A^\sigma \Lambda^\mu_\sigma g_{\mu'\nu'} \Lambda^{\nu'}_\rho A^\rho = A^\sigma g_{\sigma\rho} A^\rho$$

$$\therefore \Lambda^\sigma_\mu g_{\sigma\rho} \Lambda^\rho_\nu = g_{\mu\nu}$$

• It is obvious to see that rotations are part of Lorentz transformation

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \boxed{R} \\ 0 & & & \\ 0 & & & \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

• There are additional transformations called "boost" (having a moving observer)

Exercise

① Find the form of boost transt, by assuming that $\Lambda = \begin{pmatrix} \Lambda_{11} & \Lambda_{12} & 0 & 0 \\ \Lambda_{21} & \Lambda_{22} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ and plug back in $\Lambda^T g \Lambda = g$ then comparing with time dilation $\Delta t' = \frac{\Delta t}{\sqrt{1 - \beta^2}}$

Examples of 4-vectors in physics

• Coordinate difference dx^μ

• Velocity $u^\mu = \frac{dx^\mu}{d\tau} = (\gamma \frac{dx}{d\tau}, \gamma \vec{v})$ $\rightarrow d\tau = \frac{1}{\gamma} dt$ is Lorentz invariant

• Energy-momentum $p^\mu = mu^\mu$
 $= (\gamma \frac{mc^2}{c}, \gamma m \vec{v})$
 $= (E/c, \vec{p}) = (E, \vec{p})$

$$ds^2 = ds_\mu ds^\mu = dt^2 - dx^2 - dy^2 - dz^2 = dt^2 \left(1 - \left(\frac{d\vec{x}}{dt}\right)^2\right) \\ = dt^2 (1 - \beta^2) = dt^2 \frac{1}{\gamma^2} \equiv d\tau^2$$

• Wave vector $k^\mu = (\omega/c, \vec{k}) = (\omega, \vec{k})$

• Current $j^\mu = (\rho, \vec{j}) = (\rho, \vec{j})$

• EM potential $A^\mu = (\phi, \vec{A}) = (\phi, \vec{A})$

• Differential operator $\partial^\mu = \frac{\partial}{\partial x_\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\vec{\nabla}\right) = \left(\frac{\partial}{\partial t}, -\vec{\nabla}\right)$ $\rightarrow$ minus sign bec. this is naturally defined in covector space

Examples of scalar

• The spacetime distance $ds^2 = ds^\mu ds_\mu = dt^2 - dx^2 - dy^2 - dz^2$

• Einstein's relation $p^2 = p^\mu p_\mu = E^2 - \vec{p}^2 = m^2$

• d'Alembertian $\partial_\mu \partial^\mu = \frac{\partial^2}{\partial t^2} - \nabla^2 \equiv \square$

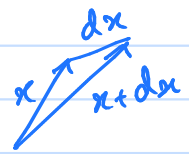
• Phase of a wave $kx = k_\mu x^\mu = \omega t - \vec{k} \cdot \vec{x}$

• Current conservation $\partial_\mu j^\mu = \frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$

• Volume element $dx^4 = dt dx dy dz$

Motion in Minkowski space

Although this notation is for components of the vectors
It is sometimes loosely called "vector"



Consider 2 events $x^\mu = (t, x, y, z)$, $x^\mu + dx^\mu = (t+dt, x+dx, y+dy, z+dz)$

$$dx^\mu = (dt, dx, dy, dz), \quad dx_\mu = (dt, -dx, -dy, -dz)$$

The spacetime distance ds^2 is defined as

$$ds^2 = dx_\mu dx^\mu = dt^2 - dx^2 - dy^2 - dz^2 = (dt \ dx \ dy \ dz) \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \begin{pmatrix} dt \\ dx \\ dy \\ dz \end{pmatrix}$$

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = g^{\mu\nu} dx_\mu dx_\nu$$

★ Can we define $g^{\mu\nu}$?

Yes! $g_{\mu\nu}$ is the metric for contravariant vectors (dx^μ and dx^μ)

$g^{\mu\nu}$ can be defined as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = g^{\mu\nu} dx_\mu dx_\nu$$

$$\Rightarrow g^{\mu\nu} g_{\mu\sigma} dx^\sigma g_{\nu\rho} dx^\rho = g_{\nu\rho} dx^\nu dx^\rho$$

$$\Rightarrow g_{\nu\rho} = g_{\sigma\mu} g^{\sigma\mu} g_{\nu\rho}$$

$$g_{\mu\nu} (g^{-1})^{\nu\alpha} = g_{\sigma\mu} g^{\sigma\beta} g_{\beta\nu} (g^{-1})^{\nu\alpha} \Rightarrow \delta_\mu^\alpha = g_{\sigma\mu} g^{\sigma\alpha}$$

$$\Rightarrow g^{\mu\nu} \text{ is the inverse of } g_{\mu\nu}$$

Proper time

As long as $ds^2 > 0$, we can find an inertial frame such that, $dx=dy=dz=0$

$$\Rightarrow ds^2 = d\tau^2 \Rightarrow d\tau \text{ is called proper time interval}$$

(It just means a clock at rest $\Rightarrow$ instantaneous rest frame)

Exercise

$$d\tau = \sqrt{1-v^2} dt = \frac{dt}{\gamma}$$

② Derive the time dilation formula from the proper time concept

③ Derive the length contraction formula by introducing similarly the proper length

4-velocity, 4-momentum

In SR, we use proper time to define 4-velocity

$$u^\mu = \frac{dx^\mu}{ds} = \frac{dx^\mu}{d\tau} = \left(\frac{dt}{d\tau}, \frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau} \right) \\ = (\gamma, \gamma v_x, \gamma v_y, \gamma v_z) = \gamma(1, \vec{v})$$

From exercise
 $d\tau = \frac{1}{\gamma} dt$

$$\therefore u_\mu u^\mu = \gamma^2(1 - |\vec{v}|^2) = \gamma^2 \left(\frac{1}{\gamma^2} \right) = 1 \quad \text{4-vels have length} = 1^*$$

Even an object at rest has $u^\mu = (1, 0, 0, 0)$

Exercise (4) Prove that $a^\mu = \frac{d u^\mu}{d\tau}$ is always orthogonal to u^μ

4-momentum is defined as $p^\mu = m u^\mu = (\gamma m, \gamma m \vec{v})$

We further call $p^\mu = (E, \vec{p})$, $E = \text{relativistic energy}$
 $\vec{p} = \text{relativistic momentum}$

(in particle physics)
we simply call these
energy & momentum

$$\Rightarrow E = \gamma m, \vec{p} = \gamma m \vec{v}$$

Exercise

(5) Show that for $|\vec{v}| \ll 1$, $E \approx m + \frac{1}{2} m v^2$

From the def. of $p^\mu = m u^\mu \Rightarrow p_\mu p^\mu = m^2$

$$\Rightarrow E^2 - |\vec{p}|^2 = m^2$$

we automatically obtain Einstein's relation

Relativistic Collisions

The conservation of energy and momentum = The conservation of 4-momentum

$$p_{\text{final}}^\mu = (E_f, \vec{p}_f), p_{\text{ini}}^\mu = (E_i, \vec{p}_i) \Rightarrow \begin{cases} E_f = E_i \\ \vec{p}_f = \vec{p}_i \end{cases}$$

From the property of 4-momentum, we have the extra equation to work with

$$\text{Let } p^\mu = p_1^\mu + p_2^\mu \Rightarrow p_\mu p^\mu = (p_1^\mu + p_2^\mu) \cdot (p_{1\mu} + p_{2\mu})$$

$\swarrow$ final particle $\uparrow$ particle 1 $\nwarrow$ particle 2

$$P_\mu P^\mu = P_{1\mu} P_1^\mu + P_{2\mu} P_2^\mu + 2 P_{1\mu} \cdot P_2^\mu$$

$$M^2 = m_1^2 + m_2^2 + 2 P_{1\mu} \cdot P_2^\mu$$

This is true for any inertial frame $\Rightarrow$ The rest frame of particle 2

$$P_1^\mu = (E_1, \vec{p}_1), \quad P_2^\mu = (m_2, \vec{0})$$

$$\Rightarrow M^2 = m_1^2 + m_2^2 + 2m_2 E_1 \Rightarrow \text{This is how we turn energy to the final mass}$$

Exercise

⑥ Work out the Compton scattering using 4-momentum, i.e., obtain

$$\Delta\lambda = \lambda' - \lambda = \frac{2\pi}{m_e}(1 - \cos\theta), \text{ by assuming that } E_{\text{photon}} = |\vec{p}|_{\text{photon}} = \frac{2\pi}{\lambda}$$

Rapidity & Pseudo-rapidity

A boost transf in x -direction $\Rightarrow \Lambda = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \beta = v$

We can reparametrise $\Lambda(\eta) = \begin{pmatrix} \cosh\eta & \sinh\eta & 0 & 0 \\ \sinh\eta & \cosh\eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

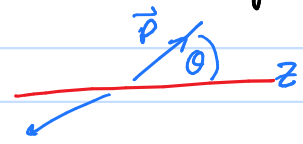
$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

Exercise ⑦ show that $\Lambda(\eta'') = \Lambda(\eta')\Lambda(\eta) \Rightarrow \eta'' = \eta' + \eta$

⑧ show that $\beta = \tanh\eta, \quad \eta = \frac{1}{2} \ln\left(\frac{1+\beta}{1-\beta}\right)$ hint: solve $\beta = \frac{e^\eta - e^{-\eta}}{e^\eta + e^{-\eta}}$

In collider physics experiment, we use pseudorapidity to describe the angle of a particle relative to the beam axis ($+z$)

$(\beta_z = v = \frac{p_z}{E})$
high energy $E \sim p$



$$\eta = \frac{1}{2} \ln\left(\frac{1+\beta_z}{1-\beta_z}\right) = \frac{1}{2} \ln\left(\frac{E+p_z}{E-p_z}\right) \cong \frac{1}{2} \ln\left(\frac{E+E\cos\theta}{E-E\cos\theta}\right) = \frac{1}{2} \ln\left(\frac{1+(\cos^2\theta_2 - \sin^2\theta_2)}{1-(\cos^2\theta_2 - \sin^2\theta_2)}\right)$$

$$= \frac{1}{2} \ln\left(\frac{2\cos^2\theta_2}{2\sin^2\theta_2}\right)$$

$$\boxed{\eta' = -\ln(\tan\theta_2)}$$

Particle Physics Homework 2

1) Rapidity

Another way of writing the boost transformation is writing velocity β in terms of rapidity η :

$$\begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cosh\eta & \sinh\eta & 0 & 0 \\ \sinh\eta & \cosh\eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

1.1). Show that $\eta = \frac{1}{2} \ln \left(\frac{1+\beta}{1-\beta} \right)$

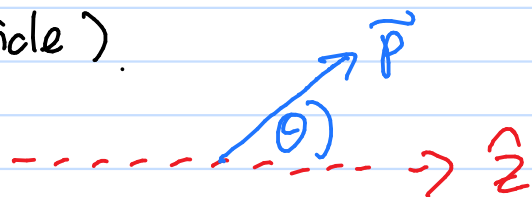
1.2). One of the advantages of using rapidity is the additional property. Show that

$$\Lambda(\eta_1) \Lambda(\eta_2) = \Lambda(\eta_1 + \eta_2)$$

1.3). In collider physics experiment, we usually deal with a very high energetic particle, i.e., its 4-momentum can be approximated as $p^\mu = (E, \vec{p}) \simeq (E, E\hat{p})$

(such that $p^\mu p_\mu \sim 0$). Consider a particle scattered off with a 3-momentum $\vec{p}$ and an angle θ with respect to $\hat{z}$ ($\hat{z}$ = direction of the incoming particle).

Show that the rapidity can



be approximated by pseudorapidity

$$\eta' = -\ln \left(\tan\left(\frac{\theta}{2}\right) \right)$$

2). 2-body decay of an unstable particle

Consider an unstable mother particle at rest with mass M . It eventually decay into 2 daughter particles with mass m_1 & m_2 respectively.

2.1) Show that the energies of daughter particles are

$$E_1 = \frac{1}{2M} (M^2 + m_1^2 - m_2^2)$$

$$E_2 = \frac{1}{2M} (M^2 + m_2^2 - m_1^2)$$

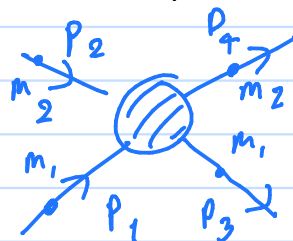
2.2) Show that the magnitude of momentum of daughter particles are

$$|\vec{p}| = \frac{1}{2M} \sqrt{(M^2 - (m_1 - m_2)^2)(M^2 - (m_1 + m_2)^2)}$$

2.3) Consider $E_1, E_2, |\vec{p}|$ in the limit that $m_1 = m_2$

3) Elastic Collision

Consider the case of elastic collision between 2 particles. The 4-momentum of 2 incident particles are p_1^μ & p_2^μ and the 4-momentum of scattered particles are p_3^μ & p_4^μ . The masses for 2 particles are m_1 & m_2 such that $p_1^\mu p_{1\mu} = p_3^\mu p_{3\mu} = m_1^2$ and $p_2^\mu p_{2\mu} = p_4^\mu p_{4\mu} = m_2^2$



There are 3 Lorentz invariants that are useful:

$$s = (p_1^\mu + p_2^\mu)(p_{1\mu} + p_{2\mu}) \equiv (p_1 + p_2)^2 = (p_3 + p_4)^2$$

$$t = (p_1^\mu - p_3^\mu)(p_{1\mu} - p_{3\mu}) \equiv (p_1 - p_3)^2 = (p_2 - p_4)^2$$

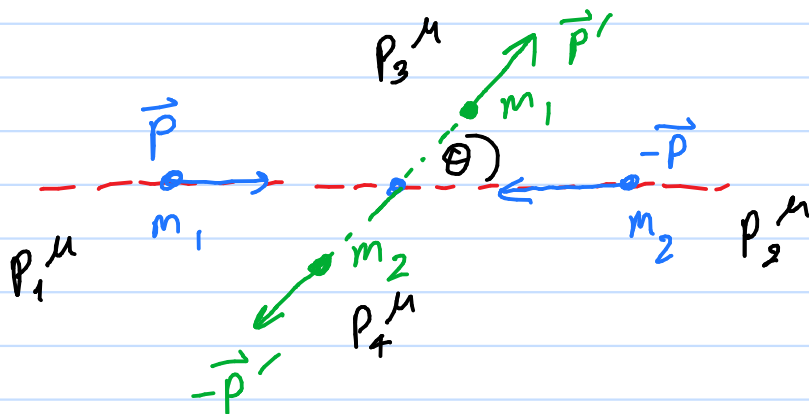
$$u = (p_1^\mu - p_4^\mu)(p_{1\mu} - p_{4\mu}) \equiv (p_1 - p_4)^2 = (p_2 - p_3)^2$$

These are called Mandelstam variables.

3.1) Show that $s + t + u = 2m_1^2 + 2m_2^2$

3.2) Let's calculate them in the center of momentum frame.

Consider $p_1^\mu = (E_1, \vec{p})$, $p_2^\mu = (E_2, -\vec{p})$



Show that $s = (E_1 + E_2)^2$. Therefore sometimes we call $\sqrt{s}$ = center of mass energy.

3.3) Show that in the center of momentum frame,

$$t = 2m_1^2 - 2E_1E_3 + 2\vec{p} \cdot \vec{p}'$$

$$u = m_1^2 + m_2^2 - 2E_1E_4 + 2\vec{p} \cdot \vec{p}'$$

3.4) Take the high energy limit ($m_1 \rightarrow 0$, $m_2 \rightarrow 0$)

Show that

$$t \approx -2E_1E_3(1 - \cos\theta)$$

Field Quantisation

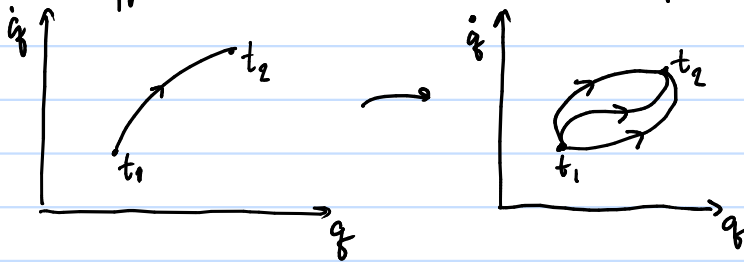
- Particle = quanta of a field
- Before quantisation, we need to understand the Lagrangian formalism on a field
- The formalism will help us understand the symmetry, charge, current (Noether's theorem)

Recap on action principle

- In classical mechanics, we can start with Newton's equation of motion $\vec{F} = m\vec{a}$
- The path of particle motion is constructed by connecting infinitesimal changes at each point

$\Rightarrow$ "The classical path of particle motion = a solution of the equation of motion"

- There is another approach based on the least action principle



The action from a point $q_1 = q(t_1)$ to $q_2 = q(t_2)$

The value of S depend on the path that the particle makes

$$S = \int_{t_1}^{t_2} dt L(q, \dot{q})$$

q is the coordinate
 $\dot{q}$ is the coor-velocity

Kinetic energy $L = K - V$

where $L(q, \dot{q}) = \frac{1}{2} m \dot{q}^2 - V(q) \rightarrow$ potential

The action principle states that the actual path is the path which minimises the action

$$\delta S = 0 = \int_{t_1}^{t_2} dt \delta L = \int_{t_1}^{t_2} dt \left[\frac{\delta L}{\delta q} \delta q + \frac{\delta L}{\delta \dot{q}} \delta \dot{q} \right]$$

we use $\delta \dot{q} = \delta \left(\frac{dq}{dt} \right) = \frac{d}{dt} (\delta q)$

and $\frac{d}{dt} (f g) = f \frac{dg}{dt} + \frac{df}{dt} g$

$$= \int_{t_1}^{t_2} dt \left[\frac{\delta L}{\delta q} \delta q + \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \delta q \right) - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) \delta q \right]$$

$$= \left[\frac{\delta L}{\delta \dot{q}} \delta q \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} dt \left[\frac{\delta L}{\delta q} - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) \right] \delta q$$

$\delta q(t_1) = \delta q(t_2) = 0$

This true for any δq
 $\Rightarrow \frac{\delta L}{\delta q} - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) = 0$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \quad \text{Euler-Lagrange equation}$$

Example

SHO: $L = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} k q^2 \Rightarrow m \ddot{q} + k q = 0 \Rightarrow \ddot{q} = -\frac{k}{m} q$

Hamiltonian Formalism

- Another equivalent way of finding e.o.m
- $L(q, \dot{q}) \Rightarrow H(q, p)$ where $p = \frac{\partial L}{\partial \dot{q}}$ (canonical conjugate momentum)
- Then consider

$$\delta L = \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} = \frac{\partial L}{\partial q} \delta q + p \delta \dot{q} = \frac{\partial L}{\partial q} \delta q + \delta(p \dot{q}) - \dot{q} \delta p$$

Let's define $H = p \dot{q} - L$ (Legendre transform)

$$\delta H = \delta(p \dot{q}) - \delta L = -\frac{\partial L}{\partial q} \delta q + \dot{q} \delta p = -\dot{p} \delta q + \dot{q} \delta p$$

We obtain

$$\dot{p} = -\frac{\partial H}{\partial q}, \quad \dot{q} = \frac{\partial H}{\partial p}$$

Example

SHO: $H = \frac{p^2}{2m} + \frac{1}{2} k q^2 \Rightarrow \dot{q} = \frac{p}{m}, \quad \dot{p} = -k q \Rightarrow m \ddot{q} = -k q$

Now let's imagine a system of N^3 particles oscillating in the lattice $L \times L \times L$, each oscillator has a volume $(\frac{L}{N})^3$

$$S = \int dt \sum_{r=1}^{N^3} L(q_{r,t}, \dot{q}_{r,t}) = \int dt \sum_{r=1}^{N^3} \left(\frac{1}{2} m_r \dot{q}_{r,t}^2 - \frac{1}{2} k_r q_{r,t}^2 \right)$$

As discussed before, we take $L/N \rightarrow 0$ & $N \rightarrow \infty \Rightarrow$ The system becomes field

The analogy: $q_{r,t} \rightarrow \phi(x)$

Define $\lim_{N \rightarrow \infty} \sum_{r=1}^{N^3} L(q_{r,t}, \dot{q}_{r,t}) = \int d^3x \mathcal{L}(\phi, \partial_\mu \phi)$

← Lagrangian density } sometimes just "Lagrangian"

Therefore, the action for a field is

$$S = \int dt d^3x \mathcal{L}(\phi, \partial_\mu \phi) \equiv \int d^4x \mathcal{L}(\phi, \partial_\mu \phi)$$

→ Now the parameters is not just t but $(t, \vec{x})$

We can apply the least action principle to find e.o.m

$$\begin{aligned} \delta S &= \int d^4x \left[\frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right] \\ &= \int d^4x \left[\frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right) - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \delta \phi \right] \\ &= \int d^4x \cancel{\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right)} + \int d^4x \left[\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \right] \delta \phi \end{aligned}$$

← true for all path
boundary terms $\int d^4x \frac{\partial}{\partial t} \Box = \Box \Big|_{t_1}^{t_2} = 0$, $\int d^4x \vec{\nabla} \Box = \Box \Big|_{x \rightarrow -\infty}^{x \rightarrow \infty} = 0$
($\delta \phi(t_1) = \delta \phi(t_2) = 0$) ($\delta \phi(x \rightarrow \infty) = 0$)

$$\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) - \frac{\delta \mathcal{L}}{\delta \phi} = 0$$

Euler-Lagrange equation

Example Klein-Gordon equation = scalar field

Consider $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 = \frac{1}{2} g_{\mu\nu} \partial^\mu \phi \partial^\nu \phi - \frac{1}{2} m^2 \phi^2$ ← kinda like kinetic term
← potential term

$$\begin{aligned} \frac{\delta \mathcal{L}}{\delta \phi} &= m^2 \phi, \quad \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} = \frac{\delta}{\delta \partial_\mu \phi} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \right) = \frac{1}{2} g^{\alpha\beta} \left(\frac{\delta \partial_\alpha \phi}{\delta \partial_\mu \phi} \right) \partial_\beta \phi + \frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \left(\frac{\delta \partial_\beta \phi}{\delta \partial_\mu \phi} \right) \\ &= \frac{1}{2} g^{\mu\beta} \partial_\beta \phi + \frac{1}{2} g^{\alpha\mu} \partial_\alpha \phi = \partial^\mu \phi \end{aligned}$$

∴ E-L equation $\Rightarrow \partial_\mu \partial^\mu \phi + m^2 \phi = 0$ Klein-Gordon equation

A generalisation is $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi) \Rightarrow \partial_\mu \partial^\mu \phi + \frac{\partial V}{\partial \phi} = 0$

Example: Maxwell's equations

$$\vec{\nabla} \cdot \vec{E} = \rho_{\text{E}}, \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \frac{\partial \vec{E}}{\partial t}, \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \vec{\nabla} \cdot \vec{B} = 0$$

Since $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$ & $\vec{\nabla} \times (-\vec{\nabla} \phi) = 0$, $\vec{E}$ & $\vec{B}$ can be written as the potentials

$$\phi \text{ & } \vec{A} \text{ where } \vec{B} = \vec{\nabla} \times \vec{A}, \quad \vec{E} = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t}$$

We can write this using 4-vector $A^\mu = (\phi, \vec{A})$, $\partial^\mu = (\partial_t, -\vec{\nabla})$

$$\left. \begin{aligned} \vec{E} &= -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} &= \vec{\nabla} \times \vec{A} \end{aligned} \right\} \Rightarrow F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \quad F^{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix}$$

where $F^{0i} = -E^i$, $F^{ij} = -\epsilon^{ijk} B^k$

Maxwell's equations $\Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Rightarrow \partial_i F^{i0} = \mu_0 \rho$ ($\mu_0 \epsilon_0 = 1$)

$\vec{\nabla} \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{j} \Rightarrow \partial_i F^{ik} + \partial_0 F^{0k} = \mu_0 j^k$

We get $\partial_\mu F^{\mu\nu} = \mu_0 j^\nu$ where $j^\nu = (\rho, \vec{j})$

Exercise 7 Show that the other two are written as

$\partial_{[\mu} F_{\nu\lambda]} = \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0$ (Bianchi identity)
exterior derivative $d^2 = 0$

Back to the Lagrangian. Now consider $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = -\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu) = \frac{1}{4} [\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\nu A_\mu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu + \partial_\nu A_\mu \partial^\nu A^\mu]$$

$$= -\frac{1}{2} [(\partial_\mu A_\nu)(\partial^\mu A^\nu) - (\partial_\mu A_\nu)(\partial^\nu A^\mu)]$$

$$\Rightarrow \frac{\delta \mathcal{L}}{\delta (\partial_\mu A_\nu)} = -\frac{1}{2} \frac{\delta}{\delta (\partial_\mu A_\nu)} [\partial_\sigma A_\rho g^{\sigma\alpha} g^{\beta\gamma} \partial_\alpha A_\beta - \partial_\sigma A_\rho g^{\sigma\alpha} g^{\beta\gamma} \partial_\alpha A_\beta]$$

$$= -\frac{1}{2} [\delta_\sigma^\mu \delta_\rho^\nu g^{\sigma\alpha} g^{\beta\gamma} \partial_\alpha A_\beta + \partial_\sigma A_\rho g^{\sigma\alpha} g^{\beta\gamma} \delta_\alpha^\mu \delta_\beta^\nu - \delta_\sigma^\mu \delta_\rho^\nu g^{\sigma\alpha} g^{\beta\gamma} \partial_\alpha A_\beta - \partial_\sigma A_\rho g^{\sigma\alpha} g^{\beta\gamma} \delta_\alpha^\mu \delta_\beta^\nu]$$

$$= -\frac{1}{2} [\partial^\mu A^\nu + \partial^\mu A^\nu - \partial^\nu A^\mu - \partial^\nu A^\mu] = -(\partial^\mu A^\nu - \partial^\nu A^\mu) = -F^{\mu\nu}$$

$\therefore$ EL equation gives $\frac{\delta \mathcal{L}}{\delta A_\nu} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial_\mu A_\nu)} \right) = 0$

$\partial_\mu F^{\mu\nu} = 0 \Rightarrow$ Maxwell equation in vacuum

We can also see that
(Using int. by part.)

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = -\frac{1}{2} (\partial_\mu A_\nu)(\partial^\mu A^\nu) + \frac{1}{2} (\partial_\mu A^\mu)^2$$

i.e., $\mathcal{L} = K - V$. (Note that the kinetic $(\frac{\partial}{\partial t})^2$ is positive)

Hamil of a field

• We can define hamiltonian density $\mathcal{H} = \int d^3x \mathcal{H}$

• The conjugate momentum of $\phi(x^\mu)$ is $\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)}$ (notice that this is not $\frac{\partial \mathcal{L}}{\partial \phi(x)}$)

• Then the Legendre transformation:

$$\mathcal{H} \equiv \pi(x) \dot{\phi}(x) - \mathcal{L} \Rightarrow \dot{\pi} = -\frac{\partial \mathcal{H}}{\partial \phi}, \quad \dot{\phi} = \frac{\partial \mathcal{H}}{\partial \pi}$$

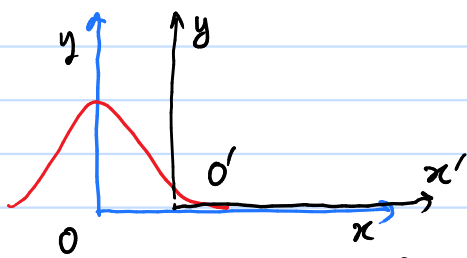
Example a scalar field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\vec{\nabla} \phi)^2 - \frac{1}{2} m^2 \phi^2$$

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi} \Rightarrow \mathcal{H} = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2$$

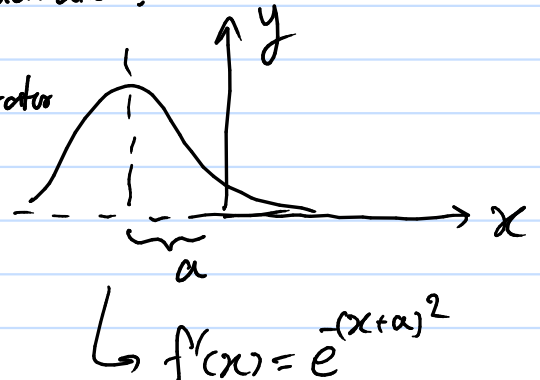
Lorentz invariance

• How do functions change under coordinate transformation?



Ex a gaussian $f(x) = e^{-x^2}$

$(x = x' + a)$ translation operator
 $x' = x - a$
 $Tx = x - a$
 $T^{-1}x = x + a$



$$f(T^{-1}x) = f'(x)$$

• The same idea can be applied to a scalar field & Lorentz transformation

• Assume Lorentz transf. $x \rightarrow x' = \Lambda x$ ($x'^\mu = \Lambda^\mu_\nu x^\nu$)

• A scalar field transforms as

$$\phi'(x) = \phi(\Lambda^{-1}x)$$

• The new field should give the same e.o.m. (same physical law)

• This can be seen from the invariance of the action $d^4x = d^4x'$
 $\& \mathcal{L} = \mathcal{L}'$

• Also, EM potential $A^\mu(x) \rightarrow \Lambda^\mu_\nu A^\nu(\bar{\Lambda}^1 x)$

• The Lorentz transformations have a "representation" on the fields

$$\phi'(x) = \phi(\bar{\Lambda}^1 x)$$

$$A'^\mu(x) = \Lambda^\mu_\nu A^\nu(\bar{\Lambda}^1 x)$$

Noether's theorem

• The connection between symmetries & conservation laws

• A symmetry of a physical law implies a conserved quantity

rotational symmetry $\Leftrightarrow$ angular momentum
conservation

$$\vec{F} \propto \vec{r} \Rightarrow \vec{\tau} = 0 \Rightarrow \vec{L} = \text{const.}$$

translational symmetry $\Leftrightarrow$ energy & momentum
(space & time) conservation

• Consider an infinitesimal change on a field

$$\phi(x) \rightarrow \phi'(x) = \phi(x) + \delta\phi(x)$$

We say that $\delta\phi = X(\phi)$ is a symmetry if the Lagrangian changes by a total derivative

$$\delta\mathcal{L} = \partial_\mu F^\mu, \text{ where } F^\mu(\phi) \text{ is a f. of fields}$$

• This means that the action is invariant (using the argument on boundary term)

• Now consider the result from any field transformation $\delta\phi$

$$\delta\mathcal{L} = \frac{\delta\mathcal{L}}{\delta\phi} \delta\phi + \frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \partial_\mu(\delta\phi)$$

$$= \left(\frac{\delta\mathcal{L}}{\delta\phi} - \partial_\mu \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \right) \right) \delta\phi + \partial_\mu \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \delta\phi \right)$$

bec. E.L eqs

$$\delta\mathcal{L} = \partial_\mu \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \delta\phi \right)$$

• For a symmetry transformation $\delta\phi = X(\phi)$, $\delta\mathcal{L} = \partial_\mu F^\mu$

• We can find an existence of the conserved current

$$\partial_\mu \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} X(\phi) \right) = \partial_\mu F^\mu$$

$$\Rightarrow \boxed{j^\mu = \frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} X(\phi) - F^\mu \quad \text{where} \quad \partial_\mu j^\mu = 0}$$

* Why do we call this a conserved current? $\Rightarrow \partial_\mu j^\mu = \frac{\partial}{\partial t} j^0 + \vec{\nabla} \cdot \vec{j} = 0$

$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{j} \rightarrow$ continuity eq^t $\Rightarrow$ The ^(global) conserved charge is $Q = \int \rho d^3x$

$$Q = \int j^0 d^3x \quad \text{such that} \quad \frac{dQ}{dt} = \int \frac{d}{dt} (j^0) d^3x = \int -\vec{\nabla} \cdot \vec{j} d^3x = 0$$

($j^0 \rightarrow 0$ as $x \rightarrow \infty$)

Translations (spacetime)

• Field $x^\mu \rightarrow x'^\mu = x^\mu - \epsilon^\mu \Rightarrow \phi(x) \rightarrow \phi'(x) = \phi(x + \epsilon^\mu) \approx \phi(x) + \epsilon^\mu \partial_\mu \phi(x)$

• Lagrangian (also scalar) $\mathcal{L}'(x) \approx \mathcal{L}(x) + \epsilon^\mu \partial_\mu \mathcal{L}(x) = \mathcal{L}(x) + \partial_\mu (\delta^\mu_\nu \epsilon^\nu \mathcal{L}(x))$

• From Noether's theorem

$$j^\mu = \frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \epsilon^\nu \partial_\nu \phi - \epsilon^\nu \delta^\mu_\nu \mathcal{L} = \epsilon^\nu \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \right)$$

$$j^\mu_\nu = \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \right) \Rightarrow \partial_\mu j^\mu_\nu = 0$$

$$T^\mu_\nu \equiv \left(\frac{\delta\mathcal{L}}{\delta(\partial_\mu\phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \right) \quad \text{the energy-momentum tensor}$$

• There are 4-currents T^μ_0, T^μ_i , 4 charges $E = \int d^3x T^0_0$, $P_i = \int d^3x T^0_i$

• E is the total energy of the field configuration / P_i is the total momentum of the field configuration

Scalar field energy-momentum tensor

- Use $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$
- we get $T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - g^{\mu\nu} \mathcal{L}$
- The conserved quantities are

$$E = \int d^3x \left(\frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 \right)$$

$$P^i = \int d^3x (\dot{\phi} \partial^i \phi)$$

- Sometimes $T^{\mu\nu}$ is not symmetric, we can add an extra term

$$T^{\mu\nu'} = T^{\mu\nu} + \partial_\rho K^{\rho\mu\nu}$$

where $K^{\rho\mu\nu}$ is some function of the fields which is anti-symmetric in $\rho\mu$

$K^{\rho\mu\nu} = -K^{\mu\rho\nu}$. This condition is there to make sure that $\partial_\mu \partial_\rho K^{\rho\mu\nu} = 0$

that is $\partial_\mu T^{\mu\nu'} = \partial_\mu T^{\mu\nu} + \partial_\mu \partial_\rho K^{\rho\mu\nu} = 0$

Exercise ① Show that the energy-momentum tensor of $\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$ is

not symmetric and is given by $T^\mu{}_\nu = -F^{\mu\rho} (\partial_\nu A_\rho) + \frac{1}{4} F^{\rho\sigma} F_{\rho\sigma} \delta^\mu{}_\nu$

② show that we can choose $T^{\mu'}{}_\nu = T^\mu{}_\nu + \partial_\lambda K^{\lambda\mu}{}_\nu$, where $K^{\lambda\mu}{}_\nu = F^{\lambda\mu} A_\nu$ to make $T^{\mu'\nu}$ symmetric.

③ show that $T^0{}_0 = \mathcal{E} = \frac{1}{2} (\vec{E}^2 + \vec{B}^2)$, $T^0{}_i = S_i = \epsilon_{ijk} E_j B_k = (\vec{E} \times \vec{B})_i$

corresponding to EM energy density and Poynting vector

Example Lorentz transformation & conserved charges

The infinitesimal form of Lorentz transformation is

$$\Lambda^\mu{}_\nu \cong \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad \text{where } \omega^\mu{}_\nu \ll 1$$

Later on we will see a more convenient form of a Lorentz transformation

$$\Lambda^\mu_\nu = e^{\frac{1}{2} \theta_{\rho\sigma} M^{\rho\sigma}} \quad \begin{matrix} \text{parameters} \\ \text{generators} \end{matrix}, \text{ where } \theta_{\rho\sigma} \text{ are antisymmetric in } \rho\sigma$$

and $M^{\rho\sigma}$ are antisymmetric in $\rho\sigma$. $M^{\rho\sigma}$ are also matrices!

$$(M^{\rho\sigma})^\mu_\nu = g^{\rho\mu} \delta^\sigma_\nu - g^{\sigma\mu} \delta^\rho_\nu$$

where

$$(M^{01})^\mu_\nu = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (M^{12})^\mu_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

↑ boost generator in x-direction
↑ rotation generator in around z-axis

Exercise ④ show that $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = e^{\theta J}$, where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

and $e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!}$ for a matrix X and $X^0 = \mathbb{I}$ for any X

⑤ show that $B(\eta) = \begin{pmatrix} \cosh\eta & \sinh\eta \\ \sinh\eta & \cosh\eta \end{pmatrix} = e^{\eta K}$, where $K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

The exponential form implies that $\omega^\mu_\nu = \frac{1}{2} \theta_{\rho\sigma} (M^{\rho\sigma})^\mu_\nu$

Let's plug back in $\Lambda^\mu_\sigma g^{\sigma\rho} \Lambda^\nu_\rho = g^{\mu\nu}$

$$(\delta^\mu_\sigma + \omega^\mu_\sigma) g^{\sigma\rho} (\delta^\rho_\nu + \omega^\rho_\nu) = g^{\mu\nu}$$

$$g^{\mu\nu} + \omega^{\mu\nu} + \omega^{\nu\mu} + O(\omega^2) = g^{\mu\nu}$$

$$\Rightarrow \omega^{\mu\nu} = -\omega^{\nu\mu} \text{ is antisymmetric}$$

One can clearly see that this is the case

$$\omega^\mu_\nu = \frac{1}{2} \theta_{\rho\sigma} (M^{\rho\sigma})^\mu_\nu = \begin{pmatrix} 0 & \theta_{01} & \theta_{02} & \theta_{03} \\ \theta_{01} & 0 & -\theta_{12} & -\theta_{13} \\ \theta_{02} & \theta_{12} & 0 & -\theta_{23} \\ \theta_{03} & \theta_{13} & \theta_{23} & 0 \end{pmatrix} \rightarrow \omega^{\mu\nu} = \begin{pmatrix} 0 & -\theta_{01} & -\theta_{02} & -\theta_{03} \\ \theta_{01} & 0 & \theta_{12} & \theta_{13} \\ \theta_{02} & -\theta_{12} & 0 & \theta_{23} \\ \theta_{03} & -\theta_{13} & -\theta_{23} & 0 \end{pmatrix}$$

Now, consider infinitesimal Lorentz transformation on a scalar field

$$\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x) = \phi(x^\mu - \omega^\mu_\nu x^\nu) = \phi(x) - \omega^\mu_\nu x^\nu \partial_\mu \phi$$

This means $\delta\phi = -\omega^\mu_\nu x^\nu \partial_\mu \phi$

$$\text{Also, } \mathcal{L} \rightarrow \mathcal{L}'; \quad \delta\mathcal{L} = -\omega^\mu_\nu x^\nu \partial_\mu \mathcal{L} = -\left[\partial_\mu (\omega^\mu_\nu x^\nu \mathcal{L}) - \underbrace{\omega^\mu_\nu \partial_\mu x^\nu \mathcal{L}}_{\omega^\mu_\nu \mathcal{L} = \omega^{\mu\nu} \partial_{\nu\mu} \mathcal{L}} \right]$$

$$= \partial_\mu (-\omega^\mu_\nu x^\nu \mathcal{L}) = \partial_\mu F^\mu$$

From Noether's theorem, the current is written as

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} (-\omega^\rho_\nu x^\nu \partial_\rho \phi) - (-\omega^\mu_\nu x^\nu \mathcal{L})$$

$$= -\omega^\rho_\nu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} x^\nu \partial_\rho \phi - \delta^\mu_\rho x^\nu \mathcal{L} \right]$$

$$= -\omega^\rho_\nu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\rho \phi - \delta^\mu_\rho \mathcal{L} \right] x^\nu = -\omega^\rho_\nu T^\mu_\rho x^\nu$$

$$= -\omega_{\rho\nu} T^{\mu\rho} x^\nu \Rightarrow \partial_\mu (T^{\mu\rho} x^\nu) = 0$$

There are 6 parameters of $\omega_{\rho\nu} \Rightarrow$ There are 6 symmetry currents/charges

We define $J^{\mu\rho\nu} \equiv T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu \Rightarrow \partial_\mu J^{\mu\rho\nu} = 0$

What are they? Rotation $\Rightarrow$ angular momentum

$$Q^{ij} = \int d^3x J^{0ij} = \int d^3x (x^i T^{0j} - x^j T^{0i}) \Rightarrow (x^i p^j - x^j p^i)$$

Boost $\Rightarrow$ centre of mass (kind of)

$$Q^{0i} = \int d^3x J^{00i} = \int d^3x (x^0 T^{0i} - x^i T^{00}) \Rightarrow (t p^i - x^i E)$$

$$\text{At } t=0, \quad Q^{0i} = \int d^3x (-\delta m x^i)$$

Internal symmetry

- So far, we have seen symmetries from coordinate transformations.
- However, there is another type of symmetries & transformations

- An internal symmetry is a symmetry from a transformation of a field
- It acts the same way at every point in spacetime $\Rightarrow$ global symmetry

Now consider a complex scalar field $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$ with a Lagrangian

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - V(\phi, \phi^*) \quad \left(\begin{array}{l} V(\phi, \phi^*) = \frac{1}{2} m^2 \phi^* \phi \\ \text{for the simplest case} \end{array} \right)$$

Exercise

(6) Show that the equations of motion are

$$\partial_\mu \partial^\mu \phi + \frac{\partial V(\phi, \phi^*)}{\partial \phi^*} = 0 \quad \text{and} \quad \partial_\mu \partial^\mu \phi^* + \frac{\partial V(\phi, \phi^*)}{\partial \phi} = 0$$

or

$$\partial_\mu \partial^\mu \phi_1 + \frac{\partial V(\phi_1, \phi_2)}{\partial \phi_1} = 0 \quad \text{and} \quad \partial_\mu \partial^\mu \phi_2 + \frac{\partial V(\phi_1, \phi_2)}{\partial \phi_2} = 0$$

• This Lagrangian has a continuous transformation:

$$\phi \rightarrow \phi' = e^{i\alpha} \phi \cong (1 + i\alpha) \phi = \phi + i\alpha \phi \Rightarrow \delta \phi = i\alpha \phi$$

$$\phi^{*'} = e^{-i\alpha} \phi^* \cong \phi^* - i\alpha \phi^* \Rightarrow \delta \phi^* = -i\alpha \phi^*$$

which leaves $\mathcal{L}$ unchanged (invariant)

$$\mathcal{L}' = \partial_\mu \phi^{*'} \partial^\mu \phi' - V(\phi^{*'}, \phi') = \partial_\mu (e^{-i\alpha} \phi^*) \partial^\mu (e^{i\alpha} \phi) - V(e^{-i\alpha} \phi^* e^{i\alpha} \phi)$$

$$= \mathcal{L} \quad \text{since} \quad \partial_\mu \alpha = 0$$

The conserved current is

$$\begin{aligned} j^\mu &= \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} \delta \phi^* \\ &= \partial^\mu \phi^* (i\alpha \phi) + \partial^\mu \phi (-i\alpha \phi^*) \end{aligned}$$

$$j^\mu \equiv i\phi \partial^\mu \phi^* - i\phi^* \partial^\mu \phi, \quad Q = \int d^3x (i\phi \partial^0 \phi^* - i\phi^* \partial^0 \phi)$$

- We will see later that this is often interpreted as electric charge / particle number (lepton number and baryon number)

Exercise (7) Consider an explicit form of a complex scalar field

$$\begin{aligned}\mathcal{L} &= \frac{1}{2} \sum_{a=1}^2 \partial_\mu \phi_a \partial^\mu \phi_a - \frac{1}{2} \sum_{a=1}^2 m^2 \phi_a^2 - g \left(\sum_{a=1}^2 \phi_a^2 \right)^2 \\ &= \frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 + \frac{1}{2} \partial_\mu \phi_2 \partial^\mu \phi_2 - \frac{1}{2} m^2 \phi_1^2 - \frac{1}{2} m^2 \phi_2^2 - g (\phi_1^2 + \phi_2^2)^2\end{aligned}$$

Instead of a rotation in the complex form $\phi \rightarrow e^{i\alpha} \phi$, we can

use $\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \rightarrow \begin{pmatrix} \phi_1' \\ \phi_2' \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$ to show that $\mathcal{L}$ is invariant

* Remark

1). One can generalise this to a rotation of N scalar fields $\phi_a \rightarrow R_a^b \phi_b$

In this case, all transformations form a "special" set of objects called

"Rotation group" $SO(N)$ [Though this is just $O(N)$ in this case]

(Special Orthogonal group)

2). One can also make another generalisation to rotate complex fields where each ϕ_a are complex. This is called $SU(N)$ group (Special Unitary)

Particle Physics Homework 3

1) EM gauge field

Consider $\mathcal{L}(A^\mu, \partial^\mu A^\nu) = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - j^\mu A_\mu$

where $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. We have learned in

the lecture that the equation of motion is

equivalent to 2 Maxwell equations.

1.1) Using $F^{\mu\nu}$ property to show the Bianchi identity

$$\partial_\mu F_{\nu\lambda} + \partial_\lambda F_{\mu\nu} + \partial_\nu F_{\lambda\mu} = 0$$

1.2) Show that the identity above is equivalent to another 2 Maxwell equations

1.3) Find the Hamiltonian of this field

2) Massive gauge field

Consider $\mathcal{L}(A^\mu, \partial^\mu A^\nu) = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + m^2 A_\mu A^\mu - j^\mu A_\mu$

where $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$

2.1) Find the equation of motion of this field

2.2) Find the Hamiltonian of this field

2.3) Show that Lorentz transformation is a symmetry of the massive gauge field, i.e.,

$$A^\mu \rightarrow A'^\mu = \Lambda^\mu_\nu A^\nu (\Lambda^{-1})^\alpha_\mu x$$

3) Energy-momentum tensor of gauge field

Consider $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$

3.1) Show that the energy momentum tensor is given by

$$T^\mu_\nu = -F^{\mu\rho} \partial_\nu A_\rho + \frac{1}{4} F^{\rho\sigma} F_{\rho\sigma} \delta^\mu_\nu$$

which is not symmetric in $\mu \leftrightarrow \nu$

3.2) Show that we can choose

$$T'^\mu_\nu = T^\mu_\nu + \partial_\lambda K^{\lambda\mu}_\nu$$

where $K^{\lambda\mu}_\nu = F^{\mu\lambda} A_\nu$

to make T'^μ_ν symmetric

3.3) Show that

$$T'^0_0 = \frac{1}{2} (\vec{E}^2 + \vec{B}^2) = \text{energy density}$$

$$T'^0_i = \epsilon_{ijk} E_j B_k = (\vec{E} \times \vec{B})_i$$

= Poynting vector

Quantisation

- In QM, we use canonical quantisation to promote Hamiltonian/Lagrangian formalism to the quantum theory

- Poisson bracket $\Rightarrow$ commutator

$$\{f(q,p), g(q,p)\}_{\text{P.B.}} \equiv \frac{\partial f}{\partial p} \frac{\partial g}{\partial q} - \frac{\partial f}{\partial q} \frac{\partial g}{\partial p} \Rightarrow [f, g] = fg - gf \quad \left(\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t} \right) \quad \text{operators}$$

$$\boxed{\begin{aligned} \{q_i, q_j\} &= \{p_i, p_j\} = 0 \\ \{q_i, p_j\} &= \delta_{ij} \end{aligned}} \quad \text{CM}$$

$\Rightarrow$

$$\boxed{\begin{aligned} [q_i, q_j] &= [p_i, p_j] = 0 \\ [q_i, p_j] &= i\delta_{ij} \end{aligned}} \quad \text{QM}$$

- From the real scalar field, we can promote ϕ & π to operators and impose commutation relations

$$[\phi(\vec{x}), \phi(\vec{y})] = [\pi(\vec{x}), \pi(\vec{y})] = 0$$

$$[\phi(\vec{x}), \pi(\vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y})$$

Note that we use an "analogy" here

$$q_r \rightarrow \phi(x)$$

- They are at the moment time independent, i.e., Schrodinger's picture (operators are fixed, states evolve)

- For a typical quantum theory, we would like to know eigenvalues/eigenvectors of the Hamiltonian. However, this is not so easy because of infinite d.o.f of a field.

(HSO at each point in space)

- We can at least solve the Hamiltonian for each point in momentum space.

- In free field, the e.o.m is

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0$$

- We take Fourier transform

$$\phi(\vec{x}, t) = \int \frac{d^3 p}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \phi(\vec{p}, t)$$

$$\phi^*(x) = \phi(x)$$

$$\int \frac{d^3 p}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \phi(\vec{p}) = \int \frac{d^3 p'}{(2\pi)^3} e^{i\vec{p}' \cdot \vec{x}} \phi(\vec{p}') \quad \downarrow \vec{p}' = -\vec{p}$$

$$\Rightarrow \phi^*(\vec{p}, t) = \phi(-\vec{p}, t)$$

$$\partial_\mu \partial^\mu \phi = (\partial_t^2 - \vec{\nabla}^2) \phi = \int \frac{d^3 p}{(2\pi)^3} \left[\frac{\partial^2 \phi(\vec{p}, t)}{\partial t^2} e^{i\vec{p} \cdot \vec{x}} - (i\vec{p}) \cdot (i\vec{p}) \phi e^{i\vec{p} \cdot \vec{x}} \right]$$

$$= \int \frac{d^3 p}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \left(\frac{\partial^2}{\partial t^2} + \vec{p}^2 \right) \phi(\vec{p}, t)$$

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0 \Rightarrow \left(\frac{\partial^2}{\partial t^2} + \vec{p}^2 + m^2 \right) \phi(\vec{p}, t) = 0$$

- It's not surprising that this resembles $\frac{d^2 x}{dt^2} = -\omega^2 x$ simple harmonic oscillators with frequency $\omega_{\vec{p}}^2 = \vec{p}^2 + m^2$ (also this looks like $p_\mu p^\mu = m^2$)

Review on SHO

- The Hamiltonian $H = \frac{1}{2} p^2 + \frac{1}{2} \omega^2 q^2$, p & q are operators $[q, p] = i$
- We use the creation & annihilation operators (raising/lowering operators) (or ladder operators)

$$a = \sqrt{\frac{\omega}{2}} q + \frac{i}{\sqrt{2\omega}} p, \quad a^\dagger = \sqrt{\frac{\omega}{2}} q - \frac{i}{\sqrt{2\omega}} p$$

$$\Rightarrow q = \frac{1}{\sqrt{2\omega}} (a + a^\dagger), \quad p = -i\sqrt{\frac{\omega}{2}} (a - a^\dagger)$$

- Substitute back to find that the commutation relation

$$[a, a^\dagger] = \left[\sqrt{\frac{\omega}{2}} q + \frac{i}{\sqrt{2\omega}} p, \sqrt{\frac{\omega}{2}} q - \frac{i}{\sqrt{2\omega}} p \right] = \frac{-i}{2} [\cancel{q}, \cancel{p}] + \frac{i}{2} [\cancel{p}, \cancel{q}] = 1$$

- The Hamiltonian is now

$$H = -\frac{1}{2} \frac{\omega}{2} (a - a^\dagger)^2 + \frac{1}{2} \frac{\omega^2}{2\omega} (a + a^\dagger)^2$$

$$= \frac{\omega}{4} \left[-(\cancel{a} - \cancel{a} a^\dagger - \cancel{a}^\dagger a + \cancel{a}^\dagger \cancel{a}^\dagger) + (\cancel{a} + \cancel{a} a^\dagger + \cancel{a}^\dagger a + \cancel{a}^\dagger \cancel{a}^\dagger) \right]$$

$$= \frac{\omega}{2} (a a^\dagger + a^\dagger a) \quad a a^\dagger - a^\dagger a = 1$$

$$= \frac{\omega}{2} (2a^\dagger a + 1) = \omega \left(a^\dagger a + \frac{1}{2} \right)$$

• Now notice that

$$[H, a^\dagger] = [\omega a^\dagger a + \cancel{\omega/2}^0, a^\dagger] = \omega (a^\dagger a a^\dagger - a^\dagger a^\dagger a) = \omega a^\dagger [a, a^\dagger] = \omega a^\dagger$$

$$[H, a] = [\omega a^\dagger a + \cancel{\omega/2}^0, a] = \omega (a^\dagger a a - a a^\dagger a) = \omega [a^\dagger, a] a = -\omega a$$

- These relations show that $a^\dagger$ & a increase/decrease energy $\Rightarrow$ going up/down between energy states
- Let a state $|E\rangle$ be an eigenstate with energy E

$$\Rightarrow H|E\rangle = E|E\rangle$$

$$H a^\dagger |E\rangle = ([H, a^\dagger] + a^\dagger H) |E\rangle = (\omega a^\dagger + a^\dagger E) |E\rangle = (E + \omega) a^\dagger |E\rangle$$

$\hookrightarrow a^\dagger |E\rangle$ is an eigenstate with energy $E + \omega$

$$H a |E\rangle = ([H, a] + a H) |E\rangle = (-\omega a + a E) |E\rangle = (E - \omega) a |E\rangle$$

$\hookrightarrow a |E\rangle$ is an eigenstate with energy $E - \omega$

- We cannot go down forever $\Rightarrow$ There must be a ground state $|0\rangle$ s.t. $a|0\rangle = 0$

which has energy $\omega/2$

$$H|0\rangle = (\omega \cancel{a^\dagger a}^0 + \omega/2) |0\rangle = \frac{\omega}{2} |0\rangle$$

- All states can then be written as

$$|n\rangle = (a^\dagger)^n |0\rangle, H|n\rangle = (n + \frac{1}{2})\omega |n\rangle$$

Now, a field theory version:

$$\phi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}} \right]$$

$$\pi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_p}{2}} \left[a_p e^{i\vec{p}\cdot\vec{x}} - a_p^\dagger e^{-i\vec{p}\cdot\vec{x}} \right]$$

As in QM version, $[\phi(\vec{x}), \phi(\vec{y})] = [\pi(\vec{x}), \pi(\vec{y})] = 0$, $[\phi(\vec{x}), \pi(\vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y})$

• They are equivalent to $[a_p, a_q] = [a_p^+, a_q^+] = 0$, $[a_p, a_q^+] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$

• This can be seen from

$$\begin{aligned}
 [\phi(\vec{x}), \pi(\vec{y})] &= \left[\int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}, \int \frac{d^3q}{(2\pi)^3} (-i) \sqrt{\frac{\omega_q}{2}} a_q e^{i\vec{q}\cdot\vec{y}} - a_q^\dagger e^{-i\vec{q}\cdot\vec{y}} \right] \\
 &= \frac{(-i)}{(2\pi)^6} \frac{1}{2} \int d^3p d^3q \sqrt{\frac{\omega_q}{\omega_p}} \left\{ [a_p, a_q] e^{i(\vec{p}\cdot\vec{x} + \vec{q}\cdot\vec{y})} - [a_p, a_q^\dagger] e^{i(\vec{p}\cdot\vec{x} - \vec{q}\cdot\vec{y})} \right. \\
 &\quad \left. + [a_p^\dagger, a_q] e^{i(\vec{q}\cdot\vec{y} - \vec{p}\cdot\vec{x})} - [a_p^\dagger, a_q^\dagger] e^{-i(\vec{p}\cdot\vec{x} + \vec{q}\cdot\vec{y})} \right\} \\
 &= \frac{(-i)}{(2\pi)^3} \frac{1}{2} \int d^3p d^3q \sqrt{\frac{\omega_q}{\omega_p}} \left[-(2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) e^{i(\vec{p}\cdot\vec{x} - \vec{q}\cdot\vec{y})} - (2\pi)^3 \delta^{(3)}(\vec{q} - \vec{p}) e^{i(\vec{q}\cdot\vec{y} - \vec{p}\cdot\vec{x})} \right] \\
 &= \frac{i}{2} \frac{1}{(2\pi)^3} \int d^3p \left(e^{i\vec{p}\cdot(\vec{x} - \vec{y})} + e^{-i\vec{p}\cdot(\vec{x} - \vec{y})} \right) \\
 &= \frac{i}{2} \left(\delta^{(3)}(\vec{x} - \vec{y}) + \delta^{(3)}(\vec{y} - \vec{x}) \right) = i \delta^{(3)}(\vec{x} - \vec{y})
 \end{aligned}$$

Note

The Dirac delta function

$$\delta^{(3)}(\vec{x} - \vec{y}) = \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\cdot(\vec{x} - \vec{y})}$$

• For Hamiltonian, one can compute H in terms of a , $a^\dagger$

$$\begin{aligned}
 H &= \frac{1}{2} \int d^3x \left(\pi^2 + (\vec{\nabla}\phi)^2 + m^2\phi^2 \right) \\
 &= \frac{1}{2} \int \frac{d^3x d^3p d^3q}{(2\pi)^6} \left[-\frac{\omega_p \omega_q}{2} (a_p e^{i\vec{p}\cdot\vec{x}} - a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}) (a_q e^{i\vec{q}\cdot\vec{x}} - a_q^\dagger e^{-i\vec{q}\cdot\vec{x}}) \right. \\
 &\quad + \frac{1}{2\sqrt{\omega_p \omega_q}} (i\vec{p} a_p e^{i\vec{p}\cdot\vec{x}} - i\vec{p} a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}) \cdot (i\vec{q} a_q e^{i\vec{q}\cdot\vec{x}} - i\vec{q} a_q^\dagger e^{-i\vec{q}\cdot\vec{x}}) \\
 &\quad \left. + \frac{m^2}{2\sqrt{\omega_p \omega_q}} (a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}) (a_q e^{i\vec{q}\cdot\vec{x}} + a_q^\dagger e^{-i\vec{q}\cdot\vec{x}}) \right] \\
 &= \frac{1}{2} \int \frac{d^3p d^3q}{(2\pi)^3} \left[-\frac{\omega_p \omega_q}{2} \left(a_p a_q \delta(\vec{p} + \vec{q}) - a_p^\dagger a_q \delta(\vec{q} - \vec{p}) - a_p a_q^\dagger \delta(\vec{p} - \vec{q}) + a_p^\dagger a_q^\dagger \delta(\vec{p} - \vec{q}) \right) \right. \\
 &\quad + \frac{1}{2\sqrt{\omega_p \omega_q}} \left(-(\vec{p} \cdot \vec{q}) a_p a_q \delta(\vec{p} + \vec{q}) - (-1)(\vec{p} \cdot \vec{q}) a_p^\dagger a_q \delta(\vec{q} - \vec{p}) - (-1)(\vec{p} \cdot \vec{q}) a_p a_q^\dagger \delta(\vec{p} - \vec{q}) + (-1)(\vec{p} \cdot \vec{q}) a_p^\dagger a_q^\dagger \delta(\vec{p} - \vec{q}) \right) \\
 &\quad \left. + \frac{1}{2\sqrt{\omega_p \omega_q}} \left(m^2 a_p a_q \delta(\vec{p} + \vec{q}) + m^2 a_p^\dagger a_q \delta(\vec{q} - \vec{p}) + m^2 a_p a_q^\dagger \delta(\vec{p} - \vec{q}) + m^2 a_p^\dagger a_q^\dagger \delta(\vec{p} - \vec{q}) \right) \right]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \int \frac{d^3 p d^3 q}{(2\pi)^6} \left[-\frac{\sqrt{\omega_p \omega_q}}{2} \left(a_{\vec{p}} a_{\vec{q}} \delta(\vec{p} + \vec{q}) - a_{\vec{p}}^\dagger a_{\vec{q}} \delta(\vec{q} - \vec{p}) - a_{\vec{p}} a_{\vec{q}}^\dagger \delta(\vec{p} - \vec{q}) + a_{\vec{p}}^\dagger a_{\vec{q}}^\dagger \delta(-\vec{p} - \vec{q}) \right) \right. \\
&\quad + \frac{1}{2\omega_p \omega_q} \left(-(\vec{p} \cdot \vec{q}) a_{\vec{p}} a_{\vec{q}} \delta(\vec{p} + \vec{q}) - (-1)(\vec{p} \cdot \vec{q}) a_{\vec{p}}^\dagger a_{\vec{q}} \delta(\vec{q} - \vec{p}) - (-1)(\vec{p} \cdot \vec{q}) a_{\vec{p}} a_{\vec{q}}^\dagger \delta(\vec{p} - \vec{q}) + (-1)(\vec{p} \cdot \vec{q}) a_{\vec{p}}^\dagger a_{\vec{q}}^\dagger \delta(-\vec{p} - \vec{q}) \right) \\
&\quad \left. + \frac{1}{2\omega_p \omega_q} \left(m^2 a_{\vec{p}} a_{\vec{q}} \delta(\vec{p} + \vec{q}) + m^2 a_{\vec{p}}^\dagger a_{\vec{q}} \delta(\vec{q} - \vec{p}) + m^2 a_{\vec{p}} a_{\vec{q}}^\dagger \delta(\vec{p} - \vec{q}) + m^2 a_{\vec{p}}^\dagger a_{\vec{q}}^\dagger \delta(-\vec{p} - \vec{q}) \right) \right] \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \left[-\frac{\omega_p}{2} \left(a_{\vec{p}} a_{-\vec{p}} - a_{\vec{p}}^\dagger a_{-\vec{p}} - a_{\vec{p}} a_{-\vec{p}}^\dagger + a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger \right) \right. \\
&\quad + \frac{1}{2\omega_p} \left(\vec{p}^2 a_{\vec{p}} a_{-\vec{p}} + \vec{p}^2 a_{\vec{p}}^\dagger a_{-\vec{p}} + \vec{p}^2 a_{\vec{p}} a_{-\vec{p}}^\dagger + \vec{p}^2 a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger \right) \\
&\quad \left. + \frac{1}{2\omega_p} \left(m^2 a_{\vec{p}} a_{-\vec{p}} + m^2 a_{\vec{p}}^\dagger a_{-\vec{p}} + m^2 a_{\vec{p}} a_{-\vec{p}}^\dagger + m^2 a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger \right) \right] \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \left[-\frac{\omega_p}{2} \left(\cancel{a_{\vec{p}} a_{-\vec{p}}} - a_{\vec{p}}^\dagger a_{-\vec{p}} - \cancel{a_{\vec{p}} a_{-\vec{p}}^\dagger} + \cancel{a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger} \right) \right. \\
&\quad \left. + \frac{\omega_p}{2} \left(\cancel{a_{\vec{p}} a_{-\vec{p}}} + a_{\vec{p}}^\dagger a_{-\vec{p}} + \cancel{a_{\vec{p}} a_{-\vec{p}}^\dagger} + \cancel{a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger} \right) \right] \\
&= \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \omega_p \left(a_{\vec{p}}^\dagger a_{-\vec{p}} + a_{\vec{p}} a_{-\vec{p}}^\dagger \right) \\
&= \int \frac{d^3 p}{(2\pi)^3} \omega_p \left(a_{\vec{p}}^\dagger a_{-\vec{p}} + \frac{(2\pi)^3}{2} \delta^{(3)}(0) \right)
\end{aligned}$$

$\delta(\vec{p} + \vec{q}) \Rightarrow \vec{q} = -\vec{p}$
 $\delta(\vec{p} - \vec{q}) \Rightarrow \vec{q} = \vec{p}$
 $\omega_{\vec{p}} = \omega_{-\vec{p}}$
 $\vec{p}^2 + m^2 = \omega_{\vec{p}}^2$

$a_{\vec{p}} a_{\vec{q}}^\dagger - a_{\vec{p}}^\dagger a_{\vec{q}} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$

- There are 2 sources of infinities, $\delta^{(3)}(0)$ & $\int \frac{d^3 p}{(2\pi)^3}$
- As in the case of ϕ we define $|0\rangle$ s.t. $a_{\vec{p}}|0\rangle = 0 \quad \forall \vec{p}$
- The energy of $|0\rangle$ is $H|0\rangle = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2} \omega_p |0\rangle$
- Quantum Field Theory always comes with ∞ . The issue is how to deal with it?
- The $\delta^{(3)}(0)$ comes from summing up all space $\delta^{(3)}(0) = \int_{-\infty}^{\infty} \frac{d^3 x}{(2\pi)^3} e^{i\vec{x}\cdot\vec{p}} \Big|_{\vec{p}=0} = V_\infty$
- large $x \rightarrow$ small $p \rightarrow$ small $\omega \rightarrow$ "infrared divergences"

- The integral for all modes of momentum also diverges

$$\int \frac{d^3p}{(2\pi)^3} (\vec{p}^2 + m^2) \rightarrow \infty \text{ bec. } |\vec{p}| \rightarrow \infty \quad (\omega \rightarrow \infty)$$

- This is called "ultraviolet divergence"
- We can deal with this term by simply subtracting it.
- This is because we only deal with energy difference (energy is measured with respect to the vacuum state)
- We redefine H as $H = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} a_{\vec{p}}^{\dagger} a_{\vec{p}}$ and $H|0\rangle = 0$
- The subtraction is nothing new. This originates from ambiguity in moving from classic to quantum. For example, we can redo everything again from the classically equivalent $H = \frac{1}{2}(\omega q - ip)(\omega q + ip) = \omega a^{\dagger} a$
- This ordering issue happens all the time
- We define "the normal ordering" of an operator ϕ as $:\phi:$ which has all annihilation operators on the right

$$\begin{aligned} \text{For example } :H: &= \int \frac{d^3p}{(2\pi)^3} \frac{\omega_{\vec{p}}}{2} : (a_{\vec{p}}^{\dagger} a_{\vec{p}} + a_{\vec{p}} a_{\vec{p}}^{\dagger}) : = \int \frac{d^3p}{(2\pi)^3} \frac{\omega_{\vec{p}}}{2} (a_{\vec{p}}^{\dagger} a_{\vec{p}} + a_{\vec{p}}^{\dagger} a_{\vec{p}}) \\ &= \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} a_{\vec{p}}^{\dagger} a_{\vec{p}} \end{aligned}$$

Particle

- Consider

$$\begin{aligned} [H, a_{\vec{p}}^{\dagger}] &= \int \frac{d^3q}{(2\pi)^3} \omega_{\vec{q}} [a_{\vec{q}}^{\dagger} a_{\vec{q}}, a_{\vec{p}}^{\dagger}] = \int \frac{d^3q}{(2\pi)^3} \omega_{\vec{q}} (a_{\vec{q}}^{\dagger} a_{\vec{q}} a_{\vec{p}}^{\dagger} - \overbrace{a_{\vec{p}}^{\dagger} a_{\vec{q}}^{\dagger} a_{\vec{q}}}^{\rightarrow}) \\ &= \int \frac{d^3q}{(2\pi)^3} \omega_{\vec{q}} a_{\vec{q}}^{\dagger} [a_{\vec{q}}, a_{\vec{p}}^{\dagger}] = \int \frac{d^3q}{(2\pi)^3} \omega_{\vec{q}} a_{\vec{q}}^{\dagger} (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \\ &= \omega_{\vec{p}} a_{\vec{p}}^{\dagger} \end{aligned}$$

and $[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$

• This means we can construct an energy state by acting $a_{\vec{p}}^\dagger$ on $|0\rangle$

• Let $|\vec{p}\rangle = a_{\vec{p}}^\dagger |0\rangle$ be a state. This has an energy

$$H|\vec{p}\rangle = H a_{\vec{p}}^\dagger |0\rangle = (a_{\vec{p}}^\dagger H + \omega_{\vec{p}} a_{\vec{p}}^\dagger) |0\rangle = \omega_{\vec{p}} |\vec{p}\rangle, \quad \omega_{\vec{p}}^2 = \vec{p}^2 + m^2$$

• Since $\omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2} = E_{\vec{p}}$, We interpret the state $|\vec{p}\rangle$ as the momentum eigenstate of a single particle of mass m .

Momentum

Take the global charge of space translation for real scalar field: $P^i = \int d^3x \dot{\phi} \partial^i \phi$

$$\vec{P} = d^3x \int \pi \vec{\nabla} \phi = - \int d^3x \int \frac{d^3p d^3q}{(2\pi)^6} (-i) \sqrt{\frac{\omega_{\vec{p}}}{2}} (a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} - a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}) \frac{1}{\sqrt{2\omega_{\vec{q}}}} (i\vec{q} a_{\vec{q}} e^{i\vec{q}\cdot\vec{x}} - i\vec{q} a_{\vec{q}}^\dagger e^{-i\vec{q}\cdot\vec{x}})$$

$$= \frac{i}{2} \int \frac{d^3p d^3q}{(2\pi)^3} q \left(i\vec{q} a_{\vec{p}} a_{\vec{q}} \delta(\vec{p} + \vec{q}) - i\vec{q} a_{\vec{p}}^\dagger a_{\vec{q}} \delta(-\vec{p} + \vec{q}) - i\vec{q} a_{\vec{p}} a_{\vec{q}}^\dagger \delta(\vec{p} - \vec{q}) + i\vec{q} a_{\vec{p}}^\dagger a_{\vec{q}}^\dagger \delta(\vec{p} - \vec{q}) \right) \sqrt{\frac{\omega_{\vec{p}}}{\omega_{\vec{q}}}}$$

$$= \frac{i}{2} \int \frac{d^3p}{(2\pi)^3} \left(-i\vec{p} a_{\vec{p}} a_{-\vec{p}} - i\vec{p} a_{\vec{p}}^\dagger a_{-\vec{p}} - i\vec{p} a_{\vec{p}} a_{\vec{p}}^\dagger - i\vec{p} a_{\vec{p}}^\dagger a_{\vec{p}}^\dagger \right)$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{\vec{p}}{2} (a_{\vec{p}} a_{-\vec{p}} + a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger + a_{\vec{p}} a_{\vec{p}}^\dagger + a_{\vec{p}}^\dagger a_{\vec{p}})$$

since $\int \frac{d^3p}{(2\pi)^3} \vec{p} a_{\vec{p}} a_{-\vec{p}} = \int \frac{d^3p}{(2\pi)^3} \vec{p} a_{-\vec{p}} a_{\vec{p}} = - \int \frac{d^3p}{(2\pi)^3} \vec{p} a_{\vec{p}} a_{-\vec{p}} = 0$

$\vec{p} \rightarrow -\vec{p}$ $[a_{\vec{p}}, a_{\vec{q}}] = 0$

and similarly for $\int \frac{d^3p}{(2\pi)^3} \vec{p} a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger = 0$

$$\Rightarrow : \vec{P} : = \int \frac{d^3p}{(2\pi)^3} \vec{p} a_{\vec{p}}^\dagger a_{\vec{p}}$$

$$\vec{P} |\vec{p}\rangle = \int \frac{d^3q}{(2\pi)^3} \vec{q} a_{\vec{q}}^\dagger a_{\vec{q}} a_{\vec{p}}^\dagger |0\rangle = \int \frac{d^3q}{(2\pi)^3} \vec{q} a_{\vec{q}}^\dagger (a_{\vec{p}}^\dagger \delta_{\vec{p}\vec{q}} + (2\pi)^3 \delta_{\vec{p}\vec{q}}^\dagger) |0\rangle = \vec{p} a_{\vec{p}}^\dagger |0\rangle$$

$\vec{p} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle$ is telling us that the state $|\vec{p}\rangle$ has momentum $\vec{p}$

Multi-Particle States

- We can create multi-particle states by applying $|\vec{p}_1, \dots, \vec{p}_n\rangle = a_{\vec{p}_1}^\dagger \dots a_{\vec{p}_n}^\dagger |0\rangle$
- Since $a^\dagger$'s are commuting $|\vec{p}, \vec{q}\rangle = |\vec{q}, \vec{p}\rangle \rightsquigarrow$ bosons
- All states form the basis for the vector space = Hilbert space.

The basis = $\{ |0\rangle, a_{\vec{p}}^\dagger |0\rangle, a_{\vec{p}}^\dagger a_{\vec{q}}^\dagger |0\rangle, \dots \}$

- This is known as Fock space = infinite sum of n-particle Hilbert space
- We can define the number operator $N = \int \frac{d^3p}{(2\pi)^3} a_{\vec{p}}^\dagger a_{\vec{p}} \Rightarrow N |\vec{p}_1, \dots, \vec{p}_n\rangle = n |\vec{p}_1, \dots, \vec{p}_n\rangle$
- It's easy to check that $[N, H] = i \frac{dN}{dt} = 0 \Rightarrow$ # of particles is conserved.
- However, interactions create and destroy particles $[N, H_{int}] \neq 0$

Normalisation

- We can see that if we normalise $\langle 0|0\rangle = 1$, we get

$$\langle \vec{p} | \vec{q} \rangle = \langle 0 | a_{\vec{p}} a_{\vec{q}}^\dagger | 0 \rangle = \langle 0 | [a_{\vec{p}}, a_{\vec{q}}^\dagger] + a_{\vec{q}}^\dagger a_{\vec{p}} | 0 \rangle = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$$
- However, this notation is not compatible with Lorentz transformation ($\delta^{(3)}$ not $\delta^{(4)}$)
- We use instead $|\vec{p}\rangle = \sqrt{2E_{\vec{p}}} a_{\vec{p}}^\dagger |0\rangle$ such that $\langle \vec{p} | \vec{q} \rangle = (2\pi)^3 \cdot 2E_{\vec{p}} \delta^{(3)}(\vec{p} - \vec{q})$
- This is Lorentz invariant since $\begin{pmatrix} p'_3 \\ E' \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} p_3 \\ E \end{pmatrix}$, $p'_3 = \gamma(p_3 + \beta E)$, $E' = \gamma(E + \beta p_3)$
- Using $\delta(f(x) - f(x_0)) = \frac{1}{|f'(x_0)|} \delta(x - x_0)$, we can compute

$$\begin{aligned} \delta^{(3)}(p'_3 - q'_3) &= \frac{1}{\left| \frac{dp'_3}{dp_3} \right|} \delta^{(3)}(p_3 - q_3) = \frac{1}{\gamma(1 + \beta \frac{dE}{dp_3})} \delta^{(3)}(p_3 - q_3) \\ &= \frac{1}{\gamma(1 + \beta \frac{E}{p_3})} \delta^{(3)}(p_3 - q_3) = \frac{E}{E'} \delta^{(3)}(p_3 - q_3) \end{aligned}$$

$E = \sqrt{p_3^2 + m^2}, \frac{dE}{dp_3} = \frac{1}{2E} \cdot 2p_3$
- $E' \delta^{(3)}(p'_3 - q'_3) = E \delta^{(3)}(p_3 - q_3)$ This is Lorentz inv.

- The completeness relation becomes

$$1 = \underbrace{\int \frac{d^3 p}{(2\pi)^3}}_{\text{Lorentz inv.}} \underbrace{\frac{1}{2E_p} |\vec{p}\rangle \langle \vec{p}|}_{\text{Lorentz inv.}}$$

which is composed of Lorentz invariant parts

- From $p_\mu p^\mu = m^2 \Rightarrow p_0^2 = E^2 = \vec{p}^2 + m^2 \Rightarrow \text{Lorentz inv.}$

- Consider the Lorentz invariant

$$\begin{aligned} \int \frac{d^4 p}{(2\pi)^4} \delta(p_0^2 - \vec{p}^2 - m^2) &= \frac{1}{(2\pi)} \int \frac{d^3 p}{(2\pi)^3} \int dp_0 \delta(p_0^2 - \vec{p}^2 - m^2) \\ &= \frac{1}{(2\pi)} \int \frac{d^3 p}{(2\pi)^3} \times \frac{1}{2\sqrt{\vec{p}^2 + m^2}} \end{aligned}$$

$= \frac{1}{2p_0} \delta(p_0 - \sqrt{\vec{p}^2 + m^2})$

$\therefore$ The measure is also Lorentz invariant.

Particle Physics Homework 4

1). Complex scalar fields

Consider a complex scalar field $\phi = \frac{\phi_1 + i\phi_2}{\sqrt{2}}$

with a Lagrangian $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi$

1.1) Find the equations of motion for ϕ & ϕ^*

1.2) In the case that $m = 0$, consider

the field translation:

$$\phi(x^\mu) \rightarrow \phi'(x^\mu) = \phi(x^\mu) + \alpha$$

where $\alpha = \text{constant}$ for any point in the spacetime.

Show that this translation is a symmetry.

1.3) From 1.2), find the Noether current & conserved charge correspond to the symmetry.

2). Gauge symmetry with gauge fixing term

Consider EM Lagrangian with a term

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\alpha} \partial_\mu A^\mu \partial_\nu A^\nu,$$

where $\alpha = \text{constant}$.

Show that the gauge transformation

$$A^\mu(x) \rightarrow A'^\mu(x) = A^\mu(x) + \partial^\mu \Lambda(x)$$

is also a symmetry of this Lagrangian

Heisenberg Picture

- Let's consider the Heisenberg picture where operators evolve
- For the operator O , we have

$$O_H = \underbrace{e^{iHt}}_{\text{Heisenberg}} O, \underbrace{e^{-iHt}}_{\text{Schrodinger}} \quad \text{and} \quad \frac{\partial O_H}{\partial t} = i[H, O_H]$$

- We label $\phi(\vec{x})$ for Schrodinger and $\phi(x^\mu) = \phi(t, \vec{x})$ for Heisenberg picture
- These two have the same commutation relation at a fixed time

$$[\phi(t, \vec{x}), \phi(t, \vec{y})] = [\pi(t, \vec{x}), \pi(t, \vec{y})] = 0$$

$$[\phi(t, \vec{x}), \pi(t, \vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y})$$

- Let's start with how ϕ, π evolve?

$$\begin{aligned} \text{From } \frac{\partial \phi}{\partial t} &= i[H, \phi] = \frac{i}{2} \int d^3y [\pi(y)^2 + (\vec{\nabla} \phi(y))^2 + m^2 \phi(y)^2, \phi(x)] \\ &= \frac{i}{2} \int d^3y [\pi(y)\pi(y), \phi(x)] + 0 + 0 \\ &= \frac{i}{2} \int d^3y \left\{ \pi(y) [\pi(y), \phi(x)] + [\pi(y), \phi(x)] \pi(y) \right\} \\ &= i \int d^3y \pi(y) (-i) \delta^{(3)}(\vec{x} - y) \\ &= \pi(x) \end{aligned}$$

$[AB, C] = ABC - CAB$
 $= ABC - ACB + ACB - CAB$
 $= A[B, C] + [A, C]B$

$$\begin{aligned} \frac{\partial \pi}{\partial t} &= i[H, \pi] = \frac{i}{2} \int d^3y [\pi(y)^2 + (\vec{\nabla} \phi(y))^2 + m^2 \phi(y)^2, \pi(x)] \\ &= 0 + \frac{i}{2} \int d^3y \left\{ \vec{\nabla} \phi(y) \cdot \vec{\nabla} [\phi(y), \pi(x)] + \vec{\nabla} [\phi(y), \pi(x)] \cdot \vec{\nabla} \phi(y) \right. \\ &\quad \left. + m^2 \phi(y) [\phi(y), \pi(x)] + m^2 [\phi(y), \pi(x)] \phi(y) \right\} \\ &= i \int d^3y \left\{ i \vec{\nabla} \phi(y) \cdot \vec{\nabla} \delta^{(3)}(y - x) + i m^2 \phi(y) \delta^{(3)}(y - x) \right\} \\ &= - \int d^3y \vec{\nabla} \phi(y) \cdot \vec{\nabla} \delta^{(3)}(y - x) - m^2 \phi(x) = \nabla^2 \phi - m^2 \phi \end{aligned}$$

by part

• Combine the 2 equations: $\vec{\nabla}^2 \phi - m^2 \phi = \frac{\partial \pi}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\partial \phi}{\partial t} \right)$

$$\Rightarrow \partial_\mu \partial^\mu \phi + m^2 \phi = 0 \quad (\text{Klein Gordon Equation})$$

• This is more relativistic

• Since $[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$ and $[H, a_{\vec{p}}^\dagger] = \omega_{\vec{p}} a_{\vec{p}}^\dagger$

• we have $H a_{\vec{p}} = a_{\vec{p}} (H - \omega_{\vec{p}})$

$$H^2 a_{\vec{p}} = H a_{\vec{p}} (H - \omega_{\vec{p}}) = a_{\vec{p}} (H - \omega_{\vec{p}})^2 \dots H^n a_{\vec{p}} = a_{\vec{p}} (H - \omega_{\vec{p}})^n$$

$$\begin{aligned} \Rightarrow e^{iHt} a_{\vec{p}} e^{-iHt} &= \left(1 + iHt + \frac{(iHt)^2}{2!} + \dots \right) a_{\vec{p}} e^{-iHt} \\ &= a_{\vec{p}} \left(1 + i(H - \omega_{\vec{p}})t + \frac{(i(H - \omega_{\vec{p}})t)^2}{2} + \dots \right) e^{-iHt} \\ &= a_{\vec{p}} e^{iHt} \cdot e^{-i\omega_{\vec{p}}t} e^{-iHt} = e^{-i\omega_{\vec{p}}t} a_{\vec{p}} \end{aligned}$$

$$\text{also } e^{iHt} a_{\vec{p}}^\dagger e^{-iHt} = e^{i\omega_{\vec{p}}t} a_{\vec{p}}^\dagger$$

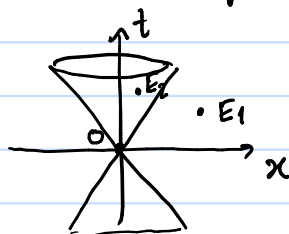
Therefore, we have a field in Heisenberg picture

$$\begin{aligned} \phi(x^\mu) &= e^{iHt} \phi(\vec{x}) e^{-iHt} = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(e^{iHt} a_{\vec{p}} e^{-iHt} \right) e^{i\vec{p} \cdot \vec{x}} + \left(e^{iHt} a_{\vec{p}}^\dagger e^{-iHt} \right) e^{-i\vec{p} \cdot \vec{x}} \\ &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(a_{\vec{p}} e^{\underbrace{-i(E_{\vec{p}}t - \vec{p} \cdot \vec{x})}_{p_\mu x^\mu}} + a_{\vec{p}}^\dagger e^{\underbrace{i(E_{\vec{p}}t - \vec{p} \cdot \vec{x})}_{p_\mu x^\mu}} \right) \end{aligned}$$

Exercise ① Check that this field is still a solution of the Klein-Gordon equation

Causality

- Any measurement between 0 and a point outside light cone is causally separated and hence equal to zero



- The commutation relation is still not fully relativistic

$$[\phi(t, \vec{x}), \pi(t, \vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y}) \Rightarrow [O_1(x^\mu), O_2(y^\mu)]$$

which must be zero outside lightcone, i.e., $[O_1(x^\mu), O_2(y^\mu)] = 0, \forall (x-y)^2 < 0$

- Let's define $\Delta(x^\mu - y^\mu) \equiv [\phi(x^\mu), \phi(y^\mu)]$

$$\begin{aligned} \Delta(x^\mu - y^\mu) &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \frac{1}{\sqrt{2E_q}} \left[a_p e^{-ipx^\mu} + a_p^\dagger e^{ipx^\mu}, a_q e^{-iqy^\mu} + a_q^\dagger e^{iqy^\mu} \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \frac{1}{\sqrt{2E_q}} \left\{ [a_p, a_q^\dagger] e^{-ipx^\mu + iqy^\mu} + [a_p^\dagger, a_q] e^{ipx^\mu - iqy^\mu} \right\} \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left(e^{-ip_\mu(x^\mu - y^\mu)} - e^{ip_\mu(x^\mu - y^\mu)} \right) \quad \left[a_p, a_q^\dagger \right] = (\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \end{aligned}$$

- This quantity is Lorentz invariant

- For time-like separation (within the lightcone) $\Delta \neq 0$

For example $x^\mu - y^\mu = (\Delta t, 0, 0, 0)$, $\Delta(x^\mu - y^\mu) = (e^{-i\Delta t} - e^{i\Delta t}) \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p}$

- But for space-like separations, $\Delta = 0$

For example $x^\mu - y^\mu = (0, \Delta x, 0, 0)$, $\Delta(x^\mu - y^\mu) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} (e^{ip_x \Delta x} - e^{-ip_x \Delta x})$

Propagators

- The amplitude of preparing a particle at point x^μ and having the particle at point y^μ is just $\langle \phi(y^\mu) | \phi(x^\mu) \rangle$

$$a_p |0\rangle = 0, \langle 0 | a_p^\dagger = 0$$

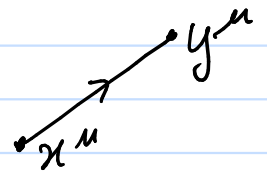
$$\begin{aligned} \langle 0 | \phi(y^\mu) \phi(x^\mu) | 0 \rangle &= \int \frac{d^3q}{(2\pi)^3} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_q}} \frac{1}{\sqrt{2E_p}} \langle 0 | (a_q e^{-iqy^\mu} + a_q^\dagger e^{iqy^\mu}) (a_p e^{-ipx^\mu} + a_p^\dagger e^{ipx^\mu}) | 0 \rangle \\ &= \int \frac{d^3q}{(2\pi)^3} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_q}} \frac{1}{\sqrt{2E_p}} \langle 0 | a_q a_p^\dagger | 0 \rangle e^{-iqy^\mu + ipx^\mu} \end{aligned}$$

Because $[a_{\vec{q}}, a_{\vec{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$, $\langle 0 | a_{\vec{q}} a_{\vec{p}}^\dagger | 0 \rangle = (2\pi)^3 \delta^{(3)}(\vec{q} - \vec{p})$ ($\langle 0 | 0 \rangle = 1$)

$$\begin{aligned} \therefore \langle 0 | \phi(y^\mu) \phi(x^\mu) | 0 \rangle &= \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 q}{4 E_p E_q} (2\pi)^3 \delta^{(3)}(\vec{q} - \vec{p}) e^{-i q_\mu y^\mu + i p_\mu x^\mu} \\ &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2 E_p} e^{-i p_\mu (y^\mu - x^\mu)} \end{aligned}$$

We define this quantity as the propagator $D(y^\mu - x^\mu)$

$$D(x-y) \equiv \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2 E_p} e^{-i p_\mu (x^\mu - y^\mu)} \equiv$$



• This function is not vanishing outside the lightcone!

$$\text{for } (x-y)^2 < 0 \Rightarrow D(x-y) \sim e^{-m|\vec{x}-\vec{y}|}$$

• This is ok since we have no measurement outside light-cone:

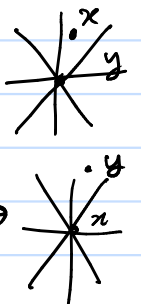
$$[\phi(x), \phi(y)] = D(x-y) - D(y-x) = 0 \text{ for } (x-y)^2 < 0$$

• This just means that the propagator $D(x-y)$ is cancelled by an equal propagator of the anti-particle moving backward.

The Feynman Propagator

• Now, a far more important quantity is the Feynman propagator

$$\Delta_F(x^\mu - y^\mu) = \langle 0 | T(\phi(x^\mu) \phi(y^\mu)) | 0 \rangle = \begin{cases} D(x^\mu - y^\mu), & x^0 > y^0 \\ D(y^\mu - x^\mu), & y^0 > x^0 \end{cases}$$



T is for time ordering operator $\Rightarrow$ reorder (earlier $\rightarrow$ right and later $\rightarrow$ left)

$$\Rightarrow T(\phi(x^\mu) \phi(y^\mu)) = \begin{cases} \phi(x^\mu) \phi(y^\mu), & x^0 > y^0 \\ \phi(y^\mu) \phi(x^\mu), & y^0 > x^0 \end{cases}$$

There is another way to write Feynman propagator by writing a 4-momentum integral

$$\Delta_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} e^{-ip_\mu(x^\mu - y^\mu)}$$

to show that this is the previous expression, we need to integrate over p^0

* we hit an infinity at $(p^0)^2 - \vec{p}^2 - m^2 = 0 \Rightarrow p^0 = \pm \sqrt{\vec{p}^2 + m^2} = \pm E_{\vec{p}}$

* To deal with this, we use complex integral (contour integral on a complex plane)

The residue theorem

Let $f(z)$ be analytic on and within a closed contour C integrated in the positive sense (counter clockwise), then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^N \text{Res}[f(z_k)]$$

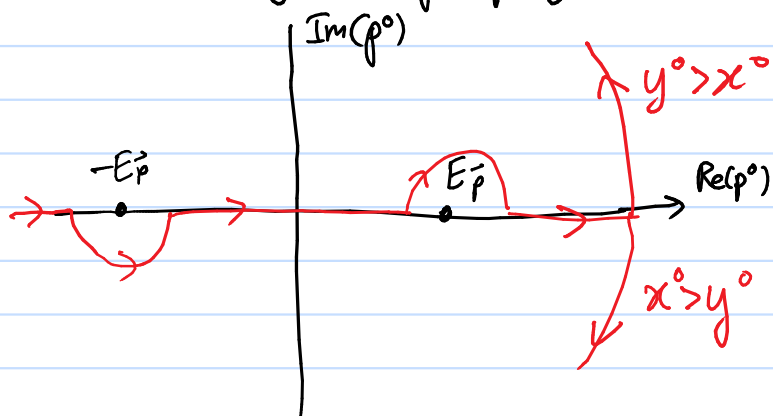
where

$$\text{Res}[f(z_k)] = \lim_{z \rightarrow z_0} [(z - z_0) f(z)]$$

where z_0 is the pole

$$\left\{ \begin{array}{l} \text{or simply look for } b_1 \text{ in} \\ f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \frac{b_1}{(z - z_0)} + \dots + \frac{b_m}{(z - z_0)^m} \end{array} \right.$$

The Feynman propagator comes from the contour



$$\int dp^0 \frac{i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_\mu(x^\mu - y^\mu)}$$

$\Rightarrow$ for $x^0 - y^0 > 0$, p^0 must be $-i\infty$ such that $e^{-ip_\mu(x^\mu - y^\mu)} \rightarrow 0$ and the contour integral goes back to line integral

$$\therefore \int dp^0 \frac{i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_\mu(x^\mu - y^\mu)} = (-2\pi i) \times \text{Res} \left(\frac{i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_\mu(x^\mu - y^\mu)} \right) \text{ for } x^0 - y^0 > 0$$

the pole inside the lower half is $p^0 = E_{\vec{p}}$

$$\text{Res} \left[\frac{i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_{\mu}(x^{\mu} - y^{\mu})} \right] = \lim_{p^0 \rightarrow E_{\vec{p}}} \frac{(p^0 - E_{\vec{p}}) i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_{\mu}(x^{\mu} - y^{\mu})}$$

$$= \lim_{p^0 \rightarrow E_{\vec{p}}} \frac{i e^{-ip_{\mu}(x^{\mu} - y^{\mu})}}{(p^0 + E_{\vec{p}})} = \frac{i}{2E_{\vec{p}}} e^{-iE_{\vec{p}}(x^0 - y^0) + i\vec{p} \cdot (\vec{x} - \vec{y})}$$

$$\therefore \int d^4 p \frac{i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_{\mu}(x^{\mu} - y^{\mu})} = \lim_{R^0 \rightarrow \infty} \oint d^4 p \frac{i}{(p^0)^2 - E_{\vec{p}}^2} e^{-ip_{\mu}(x^{\mu} - y^{\mu})}$$

$$= -2\pi i \cdot \frac{i}{2E_{\vec{p}}} e^{-iE_{\vec{p}}(x^0 - y^0) + i\vec{p} \cdot (\vec{x} - \vec{y})}$$

$$\therefore \Delta_F(x-y) = \int \frac{d^3 p}{(2\pi)^3} \cdot \frac{1}{2E_{\vec{p}}} e^{-ip_{\mu}(x^{\mu} - y^{\mu})} = D(x-y) \quad \text{for } x^0 > y^0$$

In the case of $y^0 > x^0$ we use the upper half Contour, and we get

$$\Delta_F(x-y) = \int \frac{d^3 p}{(2\pi)^3} \cdot \frac{2\pi i \cdot i}{(-2E_{\vec{p}})} e^{iE_{\vec{p}}(x^0 - y^0) + i\vec{p} \cdot (\vec{x} - \vec{y})}$$

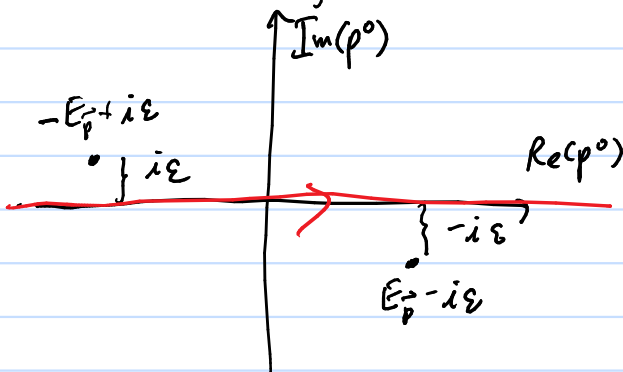
$$= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-iE_{\vec{p}}(y^0 - x^0) - i\vec{p} \cdot (\vec{x} - \vec{y})}$$

$\vec{p} \rightarrow -\vec{p}$

$$= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip_{\mu}(x^{\mu} - y^{\mu})} = D(y-x) \quad \text{for } y^0 > x^0$$

Sometimes, we choose a straight line contour on $\text{Re}(p^0)$ whereas

the poles are shifted by $\pm i\varepsilon$



$$\Delta_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i e^{-ip_{\mu}(x^{\mu} - y^{\mu})}}{p^2 - m^2 + i\varepsilon}$$

Green's functions

The propagator = Green's function for the Klein-Gordon operator

$$\begin{aligned}\Rightarrow (\partial_t^2 - \nabla^2 + m^2) \Delta_F(x-y) &= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} (-p^2 + m^2) e^{-ip \cdot (x-y)} \\ &= -i \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x-y)} \\ &= -i \delta^{(4)}(x-y)\end{aligned}$$

Interacting Fields

- What we have discussed so far are simply boring (free field/particle)
- We need interactions!
- Start by adding terms in the Lagrangian
- Consider

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \sum_{n=3}^{\infty} \frac{\lambda_n}{n!} \phi^n$$

- The coefficients λ_n are called coupling constant.
- We want small perturbations and controllable theory ($\lambda \ll 1$)

Power Counting

- We set $c = \hbar = 1$, $[c] = [\hbar] = 0$, $[X] = \text{mass dimension of } X$
- This means $[E] = [m] = [p] = 1$, $[x] = [t] = -1$
- Since $[S] = [\hbar] = 0$, $[d^4 x] = -4 \Rightarrow [\mathcal{L}] = 4$
- Now consider each term $[\partial_\mu] = 1 \Rightarrow [\phi] = 1 \Rightarrow [m] = 1$

Which means $[\lambda_n] = 4 - n$

• We can categorise λ_n into 3 types

① $[\lambda_n] > 0$ There is only 1 term here $\lambda_3, [\lambda_3] = 1, [\lambda_{3/E}] = 0$

The $\lambda_3 \phi^3$ is small when $E \gg \lambda_3$ but large at $E \ll \lambda_3$

This type of couplings is called relevant (relevant to us at low energies)

② $[\lambda_n] = 0$ this term is small when $\lambda \ll 1$

This type is called marginal

③ $[\lambda_n] < 0$ ($n \geq 5$) $\Rightarrow [\lambda_n E^{n-4}] = 0$ The coupling is small at low energies and large at high energies. This is called irrelevant.

★ Irrelevant terms cause problem at high energies (UV divergences)

Sometimes we call "non-renormalisable operators"

• Theories with non-renormalisable terms are incomplete at some scale

[G_F is irrelevant ($\frac{1}{m^2}$) and we found out that Fermi interaction is not complete

we need gauge interaction term g which is dimensionless]

Wilsonian approach

• One can view this from a theory at high energy Λ ($[\Lambda] = 1$ energy scale)

• Suppose that at a low energy $E \ll \Lambda$, physics is described by a scalar field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \sum_{n=3}^{\infty} \frac{\lambda_n}{n!} \phi^n$$

• We can relate non-renormalisable terms to a more complete theory at high energy Λ

$$\Rightarrow \lambda_n = \frac{g_n}{\Lambda^{n-4}} \left\{ \begin{array}{l} \frac{\lambda_n \phi^n}{n!} \text{ are irrelevant at } \phi \ll \Lambda \Rightarrow \left(\frac{g_n}{\Lambda^{n-4}} \frac{\phi^{n+4}}{n!} \right) \phi^4 \\ \text{and they are important at } \phi \sim \Lambda \end{array} \right.$$

• where $g_n \sim \mathcal{O}(1)$

the theory at Λ
is very difficult

• we only have access to low energy experiment $\Rightarrow$ observing these operators and figure out-

★ Notice that relev., marg., irrel. are depending on the dimension of space-time

In $n=4$; mass & kinetic terms are relevant

$\frac{\lambda \phi^4}{4!}$, $\gamma \not{\partial} \gamma^* \gamma$, gauge ints. are marginal

neutrino masses, gravitational effect are irrelevant

The interaction picture

- A useful picture in QM for describing a perturbation = interacting picture

- In S-picture, we have an evolution of states $i \frac{d}{dt} |\psi\rangle_S = H |\psi\rangle_S$

- In H-picture, we have an evolution of operators $\frac{dO_H}{dt} = i [H, O_H]$

with relations $O_H = e^{iH_0 t} O_S e^{-iH_0 t}$
 $|\psi\rangle_H = e^{iH_0 t} |\psi\rangle_S$

- The interaction picture is a hybrid. We split the Hamiltonian up as

$$H = H_0 + H_{\text{int}} \quad \left(\text{for example } H_{\text{Klein-Gordon}} + \frac{\lambda}{4!} \phi^4, \lambda \ll 1 \right)$$

- Then we let H_0 controls operators and H_{int} controls states.

- The relations between S & I-picture are

$$|\psi\rangle_I = e^{iH_0 t} |\psi\rangle_S$$

$$O_I = e^{iH_0 t} O_S e^{-iH_0 t}$$

- Since H_{int} is also an operator $\Rightarrow H_I = (H_{\text{int}})_I = e^{iH_0 t} (H_{\text{int}})_S e^{-iH_0 t}$

- Therefore, the Schrodinger equation $\Rightarrow$ equation in I-picture

$$i \frac{d}{dt} |\psi\rangle_S = H_S |\psi\rangle_S \Rightarrow i \frac{d}{dt} (e^{-iH_0 t} |\psi\rangle_I) = (H_0 + H_{\text{int}})_S e^{-iH_0 t} |\psi\rangle_I$$

$$e^{-iH_0 t} i \frac{d}{dt} |\psi\rangle_I + i(-iH_0) e^{-iH_0 t} |\psi\rangle_I = H_0 e^{-iH_0 t} |\psi\rangle_I + (H_{\text{int}})_S e^{-iH_0 t} |\psi\rangle_I$$

$$i \frac{d}{dt} |\psi\rangle_I = e^{iH_0 t} (H_{\text{int}})_S e^{-iH_0 t} |\psi\rangle_I = H_I(t) |\psi\rangle_I$$

- We would like to solve $i\hbar \frac{d}{dt} |\psi\rangle_I = H_I(t) |\psi\rangle_I$

- We assume the solution

$$|\psi(t)\rangle_I = U(t, t_0) |\psi(t_0)\rangle_I$$

where $U(t, t_0)$ is a unitary time evolution operator $\begin{cases} U(t, t) = 1 \\ U(t_1, t_2) U(t_2, t_3) = U(t_1, t_3) \end{cases}$

$$i\hbar \frac{d}{dt} |\psi(t)\rangle_I = i\hbar \frac{d}{dt} U(t, t_0) |\psi(t_0)\rangle_I = H_I(t) U(t, t_0) |\psi(t_0)\rangle_I$$

$$i\hbar \frac{d}{dt} U(t, t_0) = H_I(t) U(t, t_0)$$

- This looks like $U(t, t_0) = e^{-iH_I t}$. However, with a more careful analysis we get

$$U(t, t_0) = 1 + (-i) \int_{t_0}^t dt_1 H_I(t_1) + (-i)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H_I(t_1) H_I(t_2) + (-i)^3 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \int_{t_0}^{t_2} dt_3 H_I(t_1) H_I(t_2) H_I(t_3) + \dots$$

- We can check this by simply differentiate

$$\frac{dU}{dt} = (-i) H_I(t) + (-i)^2 H_I(t) \int_{t_0}^t dt_2 H_I(t_2) + \dots$$

$$= (-i) H_I(t) \left(1 + (-i) \int_{t_0}^t dt_2 H_I(t_2) + \dots \right) = (-i) H_I(t) U(t, t_0)$$

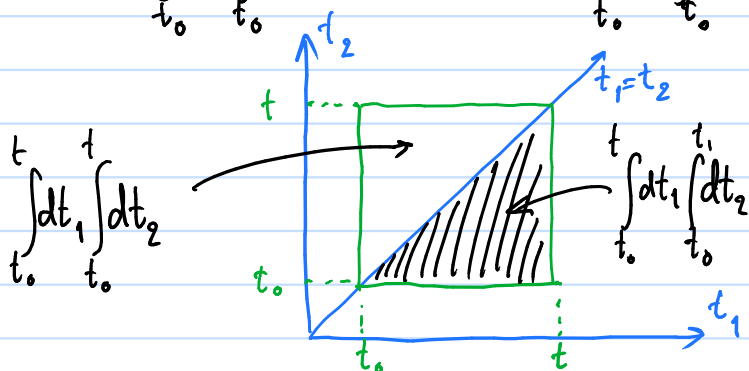
From $\int_a^b f(t) dt = F(b) - F(a)$

where $\frac{dF(t)}{dt} = f(t)$

$$\Rightarrow \frac{\partial}{\partial b} \int_a^b f(t) dt = f(b)$$

- Note all terms are in time order $\Rightarrow$ we can simplify them!

Since $\int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H_I(t_1) H_I(t_2) = \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 T \{ H_I(t_1) H_I(t_2) \}$



- Similar identity holds for the higher terms

$$\int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \dots \int_{t_0}^{t_{n-1}} dt_n H_I(t_1) \dots H_I(t_n) = \frac{1}{n!} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \dots \int_{t_0}^{t_{n-1}} dt_n T \{ H_I(t_1) \dots H_I(t_n) \}$$

$$\Rightarrow U(t, t_0) = 1 + (-i) \int_{t_0}^t dt_1 H_I(t_1) + \frac{(-i)^2}{2!} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 T \{ H_I(t_1) H_I(t_2) \} + \dots$$

$$U(t, t_0) = T \left\{ e^{-i \int_{t_0}^t dt' H(t')} \right\}$$

Heisenberg

- Another way to look at this U operator is $|\psi(t)\rangle_I = U(t, t_0) |\psi(t_0)\rangle_I$

$$\therefore |\psi(t)\rangle_I = e^{iH_0(t-t_0)} |\psi(t_0)\rangle_S = e^{iH_0(t-t_0)} e^{-iH(t-t_0)} |\psi(t_0)\rangle_H$$

$$\Rightarrow U(t, t_0) = e^{iH_0(t-t_0)} e^{-iH(t-t_0)}$$

Exercise 1 show that the field operators in H-picture & I-picture are related by

$$\phi_H(x^\mu) = U^\dagger(t, t_0) \phi_I(x^\mu) U(t, t_0)$$

- We want to study observables in the perturbation theory

- Let's define $|0\rangle$ as the vacuum in free theory and $|\Omega\rangle$ in perturbation theory

$\Rightarrow$ What we want is for example a propagator in perturbation theory

$$\Rightarrow \langle \Omega | T(\phi(x) \phi(y)) | \Omega \rangle$$

- Starting with $|0\rangle \Rightarrow$ evolve with H

$$e^{-iHT} |0\rangle = e^{-iHT} \sum_{n=0}^{\infty} |n\rangle \langle n|0\rangle = \sum_n e^{-iE_n T} |n\rangle \langle n|0\rangle$$

where E_n are the eigenvalues of H. We assume that the ground state of H: $|\Omega\rangle$

has some overlap with $|0\rangle \Rightarrow \langle \Omega | 0 \rangle \neq 0$

$$e^{-iHT} |0\rangle = e^{-iE_0 T} |\Omega\rangle \langle \Omega | 0 \rangle + \sum_{n \neq 0} e^{-iE_n T} |n\rangle \langle n|0\rangle$$

- In the limit where $T \rightarrow \infty$, the higher states are suppressed

- We have $|\Omega\rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} (e^{-iE_0 T} \langle \Omega | 0 \rangle)^{-1} e^{-iHT} |0\rangle$

- Now, we can shift T by a small constant t_0 since T is large

$$\begin{aligned} |\Omega\rangle &= \lim_{T \rightarrow \infty(1-i\epsilon)} (e^{-iE_0(T+t_0)} \langle \Omega | 0 \rangle)^{-1} e^{-iH(T+t_0)} |0\rangle \\ &= \lim_{T \rightarrow \infty(1-i\epsilon)} (e^{-iE_0(t_0 - (-T))} \langle \Omega | 0 \rangle)^{-1} e^{-iH(t_0 - (-T))} e^{iH_0(t_0 - (-T))} |0\rangle \\ &= \lim_{T \rightarrow \infty(1-i\epsilon)} (e^{-iE_0(t_0 - (-T))} \langle \Omega | 0 \rangle)^{-1} U(t_0, -T) |0\rangle \end{aligned}$$

$H_0 |0\rangle = 0, e^{iH_0 t_0} |0\rangle = |0\rangle$

Similarly we have

$$\langle \Omega | = \lim_{T \rightarrow \infty(1-i\epsilon)} \langle 0 | U(T, t_0) (e^{-iE_0(T-t_0)} \langle 0 | \Omega \rangle)^{-1}$$

- Now, we can piece everything together assuming that $x^0 > y^0 > t^0$

$$\begin{aligned} \langle \Omega | \phi(x) \phi(y) | \Omega \rangle &= \lim_{T \rightarrow \infty(1-i\epsilon)} (|\langle 0 | \Omega \rangle|^2 e^{-iE_0(2T)})^{-1} \langle 0 | U(T, t_0) U^\dagger(x^0, t_0) \phi_I(x) U(x^0, t_0) \\ &\quad \times U^\dagger(y^0, t_0) \phi_I(y) U(y^0, t_0) U(t_0, -T) |0\rangle \\ &= \lim_{T \rightarrow \infty(1-i\epsilon)} (|\langle 0 | \Omega \rangle|^2 e^{-iE_0(2T)})^{-1} \langle 0 | U(T, x^0) \phi_I(x) U(x^0, y^0) \phi_I(y) U(y^0, -T) |0\rangle \end{aligned}$$

$$\text{Since } 1 = \langle \Omega | \Omega \rangle = (|\langle 0 | \Omega \rangle|^2 e^{-iE_0(2T)})^{-1} \langle 0 | U(T, t_0) U(t_0, -T) |0\rangle$$

We have the full expression

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} \frac{\langle 0 | T \{ \phi_I(x) \phi_I(y) e^{-i \int_{-T}^T H_I(t) dt} \} |0\rangle}{\langle 0 | T \{ e^{-i \int_{-T}^T H_I(t) dt} \} |0\rangle}$$

- This tells us that we can calculate n -point f^n s in perturbation theory from calculate

$$\langle 0 | T \{ \phi \dots e^{-i \int_{-T}^T H_I(t) dt} \} |0\rangle \text{ by expanding the exponential } f^n \text{ order by order}$$

- This leads us to the problem of calculating

$$\langle 0 | T \{ \phi_I(x_1) \phi_I(x_2) \dots \phi_I(x_n) \} |0\rangle$$

Wick's theorem

- We start by a simple quantity: $\langle 0 | T \{ \phi_I(x) \phi_I(y) \} |0\rangle$

- We have already known that this is just $D_F(x-y)$

- But we will rewrite it in a form that is easy to evaluate

- We split ϕ into 2 parts

$$\phi_{\pm}(x) = \phi_{\pm}^+(x) + \phi_{\pm}^-(x) \quad \left\{ \begin{array}{l} \phi_{\pm}^+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} a_p e^{-i\vec{p}\cdot\vec{x}} \\ \phi_{\pm}^-(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} a_p^\dagger e^{i\vec{p}\cdot\vec{x}} \end{array} \right.$$

- So that $\phi_{\pm}^+(x)|0\rangle = 0$ & $\langle 0|\phi_{\pm}^-(x) = 0$

- Consider the case $x^0 > y^0$

$$\begin{aligned} T\{\phi_{\pm}(x)\phi_{\pm}(y)\} &= T\{(\phi_{\pm}^+(x) + \phi_{\pm}^-(x))(\phi_{\pm}^+(y) + \phi_{\pm}^-(y))\} \\ &= \phi_{\pm}^+(x)\phi_{\pm}^+(y) + \phi_{\pm}^+(x)\phi_{\pm}^-(y) + \phi_{\pm}^-(x)\phi_{\pm}^+(y) + \phi_{\pm}^-(x)\phi_{\pm}^-(y) \\ &= \underbrace{\phi_{\pm}^+(x)\phi_{\pm}^+(y) + \phi_{\pm}^-(y)\phi_{\pm}^+(x) + \phi_{\pm}^-(x)\phi_{\pm}^+(y) + \phi_{\pm}^-(x)\phi_{\pm}^-(y)}_{\text{all } a\text{'s are on the right / } a^\dagger\text{'s are on the left}} + [\phi_{\pm}^+(x), \phi_{\pm}^-(y)] \end{aligned}$$

all a 's are on the right / $a^\dagger$'s are on the left
These are normal order

$$T\{\phi(x)\phi(y)\}_{x^0 > y^0} = : \phi(x)\phi(y) + [\phi^+(x), \phi^-(y)] :$$

The same goes for

$$T\{\phi(x)\phi(y)\}_{y^0 > x^0} = : \phi(x)\phi(y) + [\phi^+(y), \phi^-(x)] :$$

Then we have

$$T\{\phi(x)\phi(y)\} \equiv : \phi(x)\phi(y) + \overline{\phi(x)\phi(y)} :$$

where we define the contraction as

$$\overline{\phi(x)\phi(y)} = \begin{cases} [\phi^+(x), \phi^-(y)], & x^0 > y^0 \\ [\phi^+(y), \phi^-(x)], & y^0 > x^0 \end{cases} = D_F(x-y) \xrightarrow{\text{Feynman propagator}}$$

Wick's theorem is the generalisation:

$$T\{\phi(x_1)\phi(x_2)\dots\phi(x_n)\} = : \phi(x_1)\phi(x_2)\dots\phi(x_n) + \text{all possible contractions} :$$

- For example: $(\phi(x_1) \equiv \phi_1)$

$$T\{\phi_1\phi_2\phi_3\phi_4\} = :\phi_1\phi_2\phi_3\phi_4 + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_1\phi_2} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_1\phi_3} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_1\phi_4} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_2\phi_3} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_2\phi_4} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_3\phi_4} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_1\phi_4} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_2\phi_3} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_1\phi_3} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_2\phi_4} + \overbrace{\phi_1\phi_2\phi_3\phi_4}^{\phi_3\phi_4} :$$

- Since any normal products are zero when attached with $|0\rangle$

$$\langle 0|T\{\phi_1\phi_2\phi_3\phi_4\}|0\rangle = D_F(x_1-x_2)D_F(x_3-x_4) + D_F(x_1-x_3)D_F(x_2-x_4) + D_F(x_1-x_4)D_F(x_2-x_3)$$

Exercise 2 Prove Wick's theorem by induction

Feynman diagrams

- Wick's theorem turns n-point P_n into a sum of products of Feynman propagators

- Now, we will use a diagram to represent these expressions

$$\langle 0|T\{\phi_1\phi_2\phi_3\phi_4\}|0\rangle = \begin{array}{c} 1 \\ | \\ 2 \end{array} \begin{array}{c} 3 \\ | \\ 4 \end{array} + \begin{array}{c} 1 \text{---} 3 \\ 2 \text{---} 4 \end{array} + \begin{array}{c} 1 \text{---} 4 \\ 3 \text{---} 2 \end{array}$$

- It is more interesting to consider interaction terms

$$\langle \Omega|T\{\phi(x)\phi(y)\}|\Omega\rangle \propto \langle 0|T\{\phi(x)\phi(y) + \phi(x)\phi(y)(-i\int dt H_I(t)) + \dots\}|0\rangle$$

The first term is just $D_F(x-y)$

- We use $H_I = \int d^3z \frac{\lambda}{4!} \phi^4$ as an example. Consider the 1st order in the expansion

$$\begin{aligned} \langle 0|T\{\phi(x)\phi(y)(-i)\int dt \int d^3z \frac{\lambda}{4!} \phi^4\}|0\rangle &= \langle 0|T\{\phi(x)\phi(y)(-i\frac{\lambda}{4!})\int d^4z \phi(z)\phi(z)\phi(z)\phi(z)\}\rangle \\ &= (-i\frac{\lambda}{4!})\int d^4z \langle 0|:\phi(x)\phi(y)\phi(z)\phi(z)\phi(z)\phi(z): \end{aligned}$$

$$+ \overbrace{\phi(x)\phi(y)\phi(z)\phi(z)\phi(z)\phi(z)}^{\text{not so relevant}} + \dots$$

$$+ 3\overbrace{\phi(x)\phi(y)\phi(z)\phi(z)\phi(z)\phi(z)}^{\text{3 ways to contract 4 } \phi\text{'s (which are all equal)}}$$

$$+ 4 \times 3 \times \overbrace{\phi(x)\phi(y)\phi(z)\phi(z)\phi(z)\phi(z)}^{\text{4} \times \text{3 ways to contract}}$$

$$\langle 0 | T \{ \phi(x) \phi(y) (-i) \int d^4z \frac{\lambda}{4!} \phi^4 | 0 \rangle = 3 \cdot \left(\frac{-i\lambda}{4!} \right) D_F(x-y) \int d^4z D_F(z-x) D_F(z-y) \\ + 4 \times 3 \times \left(\frac{-i\lambda}{4!} \right) \int d^4z D(x-z) D(y-z) D_F(z-z)$$

which can be expressed by Feynman diagrams

$$= 3 \times \left(\text{diagram: line from } x \text{ to } y \text{ with a loop at } z \right) + 4 \times 3 \times \left(\text{diagram: line from } x \text{ to } z \text{ and } z \text{ to } y \text{ with a loop at } z \right)$$

Exercise 3 Write Feynman diagrams for λ^2 terms in the expansion

Then we can summarise all terms into diagrams

$$\langle 0 | T \{ \phi_I(x) \phi_I(y) e^{-i \int d^4z H_I(z)} \} | 0 \rangle = \left(\text{Sum of all possible diagrams with two external points} \right)$$

The rules for associating analytic expressions with pieces of diagrams are called Feynman rules

ϕ^4 theory:

1). For each propagator $x \text{ --- } y = D_F(x-y)$

2). For each vertex $\text{X} \text{ at } z = (-i\lambda) \int d^4z$

3). For each external point $x \text{ --- } = 1$

4). Divide by the symmetry factor * (We don't have time to discuss this please do a bit of self study)

This is position-space Feynman rules. However sometimes, it is easier to do this

in momentum space by using $D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)}$

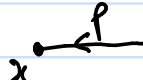
and the fact that at a vertex the momentum is conserved

$\text{diagram: vertex with four external lines labeled } p_1, p_2, p_3, p_4 \rightarrow \int d^4z e^{-ip_1 \cdot z} e^{-ip_2 \cdot z} e^{-ip_3 \cdot z} e^{+ip_4 \cdot z} = (2\pi)^4 \delta^{(4)}(\vec{p}_1 + \vec{p}_2 + \vec{p}_3 - \vec{p}_4)$

- We arrive at the momentum-space Feynman rules

1). For each propagator  $= \frac{i}{p^2 - m^2 + i\epsilon}$

2). For each vertex  $= -i\lambda$

3). For each external point  $= e^{-ip \cdot x}$

4). Impose momentum conservation at each vertex

5). Integrate over each undetermined momentum $\int \frac{d^4 p}{(2\pi)^4}$

6). Divide by the symmetry factor

* There are lots of disconnected diagrams produced this way

• However, the factor $\langle 0 | T \{ e^{i \int \mathcal{H}_I} \} | 0 \rangle$ cancel all of them!

$$\therefore \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \left(\text{Sum of all connected diagrams with } n \text{ external points} \right)$$

Cross sections & the S-Matrix

• We know how to compute n-point correlation functions using Wick's theorem and Feynman diagram technique.

• How are they related to the physical quantities in particle physics?

Cross section

• For particle physics experiment, we are interested in colliding particles at each others and count how many are coming out at a specific state (final state) (energy, momentum, type of particles, etc.)

$$\text{Number of events} = \sigma \times (\text{collider parameters})$$

- σ is called cross section, has the unit of area representing physics.

- The factor of collider factors is usually known as Luminosity, L .

(coll. factors = bunch length, density, number, etc...)

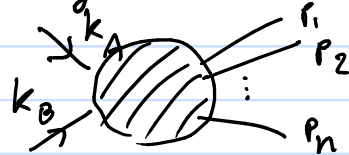
- As in quantum mechanics, # of events $\propto (\text{Prob})^2$

$$\Rightarrow \sigma \propto \text{Prob}$$

$$\sigma \propto |\langle \text{final state} | \text{initial state} \rangle|^2$$

- To calculate the overlap, we need to bring states at the same time.

- The states are labeled by momenta:



- Then we use the time evolution operator to relate them

$$\begin{aligned} \lim_{T \rightarrow \infty} \langle p_1 p_2 \dots | k_A k_B \rangle_{\text{initial}} &= \lim_{T \rightarrow \infty} \langle p_1 p_2 \dots | U(T, -T) | k_A k_B \rangle \\ &\equiv \langle p_1 p_2 \dots | S | k_A k_B \rangle \end{aligned}$$

- The states in the final line are considered at a common time.

- The unitary operator S is known as S -matrix (Scattering matrix)

- The reason for setting $T \rightarrow \infty$ is because we believe that states

at $T \rightarrow \infty, -\infty$ belong to the free theory (we know the solution)

$$\langle p_1 p_2 \dots | S | k_A k_B \rangle = \lim_{T \rightarrow \infty} \langle p_1 p_2 \dots | T(e^{-i \int_{-T}^T dt H_I^{(ct)}}) | k_A k_B \rangle$$

- We don't care too much about the trivial part of the unitary

operator $\Rightarrow S = 1 + iT$

- We have

$$\langle p_1 p_2 \dots | iT | k_A k_B \rangle = \left(\lim_{T \rightarrow \infty} \langle p_1 p_2 \dots | T(e^{-i \int_{-T}^T dt H_I^{(ct)}}) | k_A k_B \rangle \right) \text{From connected \& amputated diagrams}$$

- We will explain connected & amputated diagrams later.

- S & T should contain the 4-momentum conservation factor $\delta^{(4)}(k_A + k_B - \sum_i p_i)$

- Extracting this factor, we define the invariant matrix element M by

$$\langle p_1 p_2 \dots | iT | k_A k_B \rangle = (2\pi)^4 \delta^{(4)}(k_A + k_B - \sum_i p_i) \cdot iM(k_A k_B \rightarrow p_1 p_2 \dots)$$

- The matrix element M only depends on physics of interaction

(Other parts depend on kinematic (energy & momentum))

Example 2 → 2 scattering (ϕ^4 theory)

Let's first calculate a few terms of

$$\langle p_1 p_2 | T \left(e^{-i \frac{\lambda}{4!} \int d^4x \phi_I^4(x)} \right) | p_A p_B \rangle$$

$$= \langle p_1 p_2 | p_A p_B \rangle + \langle p_1 p_2 | T \left(-i \frac{\lambda}{4!} \int d^4x \phi_I^4(x) \right) | p_A p_B \rangle + \dots$$

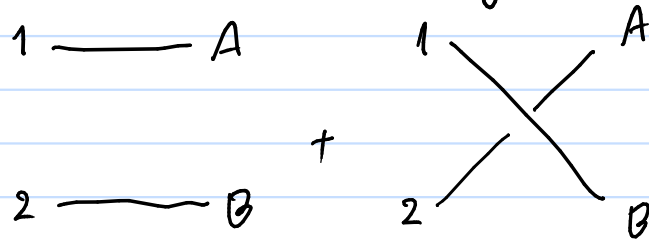
- The first term is

$$\begin{aligned} \langle p_1 p_2 | p_A p_B \rangle &= \sqrt{2E_1} \sqrt{2E_2} \sqrt{2E_A} \sqrt{2E_B} \langle 0 | a_1 a_2 a_A^\dagger a_B^\dagger | 0 \rangle \\ &= \sqrt{2E_1} \sqrt{2E_2} \sqrt{2E_A} \sqrt{2E_B} \left(\langle 0 | a_1 a_A^\dagger a_2 a_B^\dagger | 0 \rangle \right. \\ &\quad \left. + [a_2, a_A^\dagger] \langle 0 | a_1 a_B^\dagger | 0 \rangle \right) \\ &= \sqrt{2E_1} \sqrt{2E_2} \sqrt{2E_A} \sqrt{2E_B} \left(\langle 0 | a_A a_1 a_2 a_B^\dagger | 0 \rangle \right. \\ &\quad \left. + [a_1, a_A^\dagger] \langle 0 | a_2 a_B^\dagger | 0 \rangle \right. \\ &\quad \left. + [a_2, a_A^\dagger] \langle 0 | a_1 a_B^\dagger | 0 \rangle \right) \\ &= \sqrt{2E_1} \sqrt{2E_2} \sqrt{2E_A} \sqrt{2E_B} \times \left[(2\pi)^6 \delta(p_1 - p_A) \delta(p_2 - p_B) \right. \\ &\quad \left. + (2\pi)^6 \delta(p_2 - p_A) \delta(p_1 - p_B) \right] \end{aligned}$$

This part tells us that the final state is identical to the initial state

⇒ This is not a part of the matrix element.

- We can interpret this term as diagrams



- The next term can be calculated from Wick's theorem

$$\langle p_1 p_2 | T \left(-i \frac{\lambda}{4!} \int d^4 x \phi_I^4(x) \right) | p_A p_B \rangle = \langle p_1 p_2 | N \left(-i \frac{\lambda}{4!} \int d^4 x \phi_I^4(x) + \text{contractions} \right) | p_A p_B \rangle$$

- Now consider the contraction between ϕ & state

$$\begin{aligned} \phi_I^+(x) | p \rangle &= \int \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_k}} a_k e^{-ik \cdot x} \sqrt{2E_p} a_p^\dagger | 0 \rangle \\ &= \int \frac{d^3 k}{(2\pi)^3} \frac{\sqrt{2E_p}}{\sqrt{2E_k}} e^{-ik \cdot x} (2\pi)^3 \delta(k-p) | 0 \rangle \\ &= e^{-ip \cdot x} | 0 \rangle \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} a_k a_p^\dagger = [a_k, a_p^\dagger] + a_p^\dagger a_k$$

- Similarly $\langle p | \phi_I^-(x) = e^{ip \cdot x}$

- It is then natural to define

$$\overline{\phi_I^-(x) | p \rangle} = e^{-ip \cdot x} \quad \overline{\langle p | \phi_I^-(x)} = e^{ip \cdot x}$$

- They can be represented by external lines of Feynman diagrams

$$\overline{\phi_I^-(x) | p \rangle} = \text{line with arrow pointing right, labeled } p \text{ at the start}, \quad \overline{\langle p | \phi_I^-(x)} = \text{line with arrow pointing right, labeled } p \text{ at the end}$$

- With this additional rule, only fully contracted terms (ϕ & p) survive. $\Rightarrow$ We only need to calculate a few terms

- Consider

$$-i \frac{\lambda}{4!} \int d^4 x \langle p_1 p_2 | \phi \phi \phi \phi | p_A p_B \rangle = 8 \times \text{diagram with two external lines labeled 1 and 2, and two internal lines labeled A and B}$$

- This is another trivial part $\Rightarrow$ ignore this

- The on non-trivial part is

$$\frac{-i\lambda}{4!} \int d^4x \langle p_1 p_2 | \overbrace{\phi\phi\phi\phi}^{4 \text{ ways}} | p_A p_B \rangle = 4! \left(\frac{-i\lambda}{4!} \right) \int d^4x e^{-i(p_A + p_B - p_1 - p_2)x}$$

$$= -i\lambda (2\pi)^4 \delta^{(4)}(p_A + p_B - p_1 - p_2)$$

- If we compare this term with the invariant matrix element

$$i(2\pi)^4 \delta^{(4)}(p_A + p_B - p_1 - p_2) \mathcal{M} \Rightarrow \mathcal{M} = -\lambda$$

- The total cross-section would be

$$\sigma \propto |\mathcal{M}|^2 \propto \lambda^2$$

- By dimensional analysis $\sigma = \frac{k\lambda^2}{s} \left(\sigma = \frac{\lambda^2}{32\pi E_{cm}^2} \right)$
(The unit of σ is GeV^{-2})

- The conclusion is we can calculate the matrix element by constructing Feynman diagrams & translate them to $\mathcal{M}$ using Feynman rules.

- Why connected & amputated diagrams?

$\Rightarrow$ disconnected diagrams have a part of diagram that is separated from the external legs. $\Rightarrow$ This doesn't contribute to the matrix element ($S = 1 + iT \Rightarrow$ which represents interaction)

$\Rightarrow$ amputated diagrams lead to infinity $\Rightarrow$ we ignore this

$$\longrightarrow = \longrightarrow + \text{loop} + \text{2 loops} + \dots$$

Particle Physics Homework 5

1). Show that for a scalar field theory

$$[\hat{H}, \hat{a}_{\vec{p}}] = -\omega_{\vec{p}} \hat{a}_{\vec{p}}.$$

2). Show that for a scalar field theory

$$[\hat{H}, \hat{p}] = 0.$$

Does this make sense?

3). Calculate

$$\hat{H}(\hat{a}_{\vec{p}}^{\dagger} | 0 \rangle),$$

and justify your result based on previous question

4). Using the BCH formula:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \underbrace{[\hat{A}, [\hat{A}, \hat{B}]]}_{2!} + \underbrace{[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]]}_{3!} + \dots$$

to calculate

$$e^{i\hat{H}t} \hat{a}_{\vec{p}} e^{-i\hat{H}t}.$$

5). Calculate

$$\langle 0 | \hat{\pi}_H(x^\mu) \hat{\pi}_H(y^\mu) | 0 \rangle$$

Particle Physics Homework 6

1). Show that the Green f_n :

$$G(x^\mu - y^\mu) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p_\mu p^\mu} e^{-i p_\mu (x^\mu - y^\mu)}$$

corresponds to the equation:

$$\partial_\mu \partial^\mu \phi(x^\mu) = 0$$

2). Calculate the Feynman propagator

$$\Delta_F(x^\mu - y^\mu) = \int \frac{d^3 \vec{p}}{(2\pi)^3} \int dp^0 \frac{i}{(p^0)^2 - (\vec{p})^2 - m^2 + i\epsilon} e^{-i p_\mu (x^\mu - y^\mu)}$$

for $y^0 > x^0$ using the Upper Half Plane (UHP)

3). Discuss what would happen if we write

$$\Delta_F'(x^\mu - y^\mu) = \int \frac{d^3 \vec{p}}{(2\pi)^3} \int dp^0 \frac{i}{(p^0)^2 - (\vec{p})^2 - m^2 - i\epsilon} e^{-i p_\mu (x^\mu - y^\mu)}$$

instead?

4). Find the mass dimension of the field & parameters in the Lagrangian

$$4.1) \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - m A^\mu \phi \partial_\mu \phi - \lambda^\mu \partial_\mu \phi$$

$$4.2) \mathcal{L} = \frac{1}{2} \gamma^\mu \gamma \partial_\mu \gamma - \frac{1}{2} m \gamma \gamma - \lambda \gamma \gamma \gamma \gamma$$

$\uparrow$
 $[\gamma^\mu] = 0$

Particle Physics Homework 7

- 1). Find the interacting Hamiltonian in the interaction picture ($H_I = e^{iH_0 t} H_{int} e^{-iH_0 t}$) of the Lagrangian

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{3!} \phi^3$$

- 2). Show that

$$\langle \Omega | = \lim_{T \rightarrow \infty (1-i\epsilon)} \langle 0 | \hat{U}(T, t_0) \left(e^{-iE_0(T-t_0)} \langle \Omega | 0 \rangle \right)^{-1}$$

- 3). Calculate the following quantities for

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4 \quad (\phi^4\text{-theory})$$

up to the term with λ^2

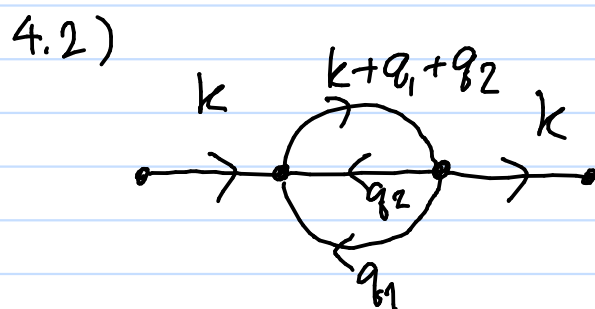
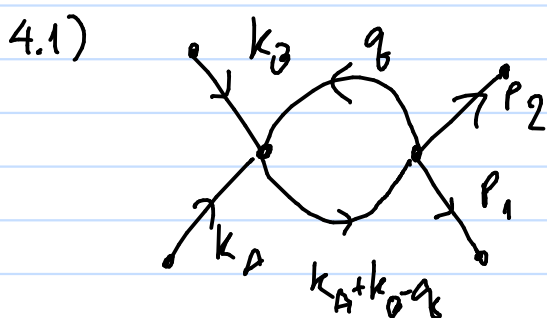
3.1) $\langle \Omega | \hat{T}(\phi(x_1)) | \Omega \rangle$

3.2) $\langle \Omega | \hat{T}(\phi(x_1^{\mu}) \phi(x_2^{\mu})) | \Omega \rangle$

3.3) $\langle \Omega | \hat{T}(\phi(x_1^{\mu}) \phi(x_2^{\mu}) \phi(x_3^{\mu})) | \Omega \rangle$

3.4) $\langle \Omega | \hat{T}(\phi(x_1^{\mu}) \phi(x_2^{\mu}) \phi(x_3^{\mu}) \phi(x_4^{\mu})) | \Omega \rangle$

- 4). Write the expression for the following diagrams in ϕ^4 theory (external lines are physical particles)



5). ϕ^3 -theory

Consider the Lagrangian

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{3!} \phi^3$$

5.1). What is the dimension of $[\lambda]$

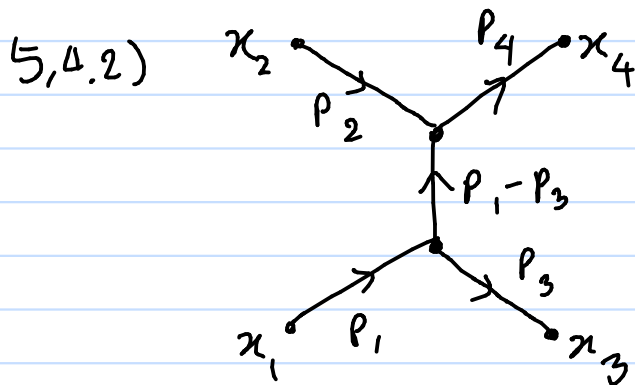
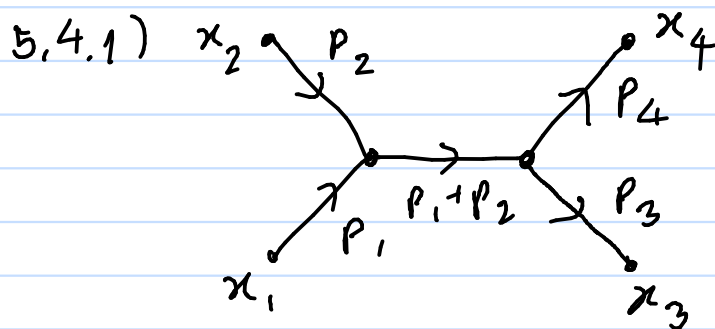
5.2). Find $\hat{T}_I(t)$

5.3). Find the following:

5.3.1). $\langle \Omega | \hat{T}(\hat{\phi}_I(x_1) \hat{\phi}_I(x_2) \hat{\phi}_I(x_3)) | \Omega \rangle$

5.3.2). $\langle \Omega | \hat{T}(\hat{\phi}_I(x_1) \hat{\phi}_I(x_2) \hat{\phi}_I(x_3) \hat{\phi}_I(x_4)) | \Omega \rangle$

5.4). Write the expression for the following diagrams
(external lines are physical particles)



The Dirac Equation

- We have learnt that a scalar field under a Lorentz transformation

$$\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x)$$

- For a vector field

$$A^\mu(x) \rightarrow \Lambda^\mu_\nu A^\nu(\Lambda^{-1}x)$$

- In a general case a field can transform as

$$\phi^a(x) \rightarrow D[\Lambda]^a_b \phi^b(\Lambda^{-1}x)$$

where the matrices $D[\Lambda]$ form a representation of the Lorentz group

$$\Rightarrow D[\Lambda_1]D[\Lambda_2] = D[\Lambda_1\Lambda_2] \quad (\text{homomorphic mapping})$$

$$\text{and } D[\Lambda^{-1}] = D[\Lambda]^{-1} \text{ \& } D[1] = 1$$

The spinor representation

- We start by defining the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}\mathbb{1}, \quad g^{\mu\nu} = \text{diag}(+, -, -, -)$$

where γ^μ with $\mu=0,1,2,3$ are a set of four matrices

- For example, in Weyl basis (or chiral basis)

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad \text{where each elements are } 2 \times 2 \text{ matrices}$$

$$\text{and } \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Now consider

$$S^{\rho\sigma} = \frac{1}{4} [\gamma^\rho, \gamma^\sigma] \begin{cases} 0, & \rho=\sigma \\ \frac{1}{2}\gamma^\rho\gamma^\sigma, & \rho \neq \sigma \end{cases} = \frac{1}{2}\gamma^\rho\gamma^\sigma - \frac{1}{2}g^{\rho\sigma}\mathbb{1}$$

- Now, consider

$$D[\Lambda] = \exp\left(\frac{1}{2}\omega_{\rho\sigma}S^{\rho\sigma}\right) \quad \left(\begin{array}{l} \text{Similar to what we learned} \\ \text{before } \Lambda = e^{i\omega_{\mu\nu}M^{\mu\nu}} \end{array} \right)$$

- However the matrix $S^{\rho\sigma}$ is not anti-hermitian, since $\gamma^{0\dagger} = \gamma^0$, $\gamma^i\dagger = -\gamma^i$

$$\Rightarrow (\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$$

- The term $\psi^\dagger \psi$ is not invariant under Lorentz transformation

$$\begin{aligned} \text{But we have } (S^{\mu\nu})^\dagger &= \frac{1}{4}[(\gamma^\nu)^\dagger, (\gamma^\mu)^\dagger] = \gamma^0 \frac{1}{4}[(\gamma^\nu)^\dagger, (\gamma^\mu)^\dagger] \gamma^0 \\ &= -\gamma^0 S^{\mu\nu} \gamma^0 \end{aligned}$$

$$\begin{aligned} \Rightarrow D[\Lambda]^\dagger &= \exp\left(\frac{1}{2}\omega_{\rho\sigma}S^{\rho\sigma\dagger}\right) = \exp\left(-\gamma^0\left(\frac{1}{2}\omega_{\rho\sigma}S^{\rho\sigma}\right)\gamma^0\right) \\ &= \gamma^0 D[\Lambda]^{-1} \gamma^0 \end{aligned}$$

- Let's define the Dirac adjoint

$$\bar{\psi}(x) = \psi^\dagger(x) \gamma^0$$

- We can show that $\bar{\psi}(x)\psi(x)$ is an invariant term

$$\bar{\psi}(x)\psi(x) = \psi^\dagger(x) \gamma^0 \psi(x)$$

$$\text{under Lorentz} \rightarrow \psi^\dagger(\Lambda^{-1}x) S[\Lambda]^\dagger \gamma^0 S[\Lambda] \psi(\Lambda^{-1}x)$$

$$= \psi^\dagger(\Lambda^{-1}x) \gamma^0 S[\Lambda]^{-1} \gamma^0 S[\Lambda] \psi(\Lambda^{-1}x)$$

$$= \psi^\dagger(\Lambda^{-1}x) \gamma^0 \psi(\Lambda^{-1}x)$$

$$= \bar{\psi}(\Lambda^{-1}x) \psi(\Lambda^{-1}x) \quad (\text{scalar transformation})$$

- Also, $\bar{\psi} \gamma^\mu \psi$ transform as a vector under Lorentz transf, i.e.,

$$\bar{\psi}(x) \gamma^\mu \psi(x) = \Lambda^\mu_\nu \bar{\psi}(\Lambda^{-1}x) \gamma^\nu \psi(\Lambda^{-1}x)$$

The proof is following:

$$\begin{aligned}\bar{\psi} \gamma^\mu \psi &\rightarrow \psi^\dagger S[\Lambda]^\dagger \gamma^\mu S[\Lambda] \psi \\ &= \psi^\dagger \gamma^\mu S[\Lambda]^{-1} \gamma^\mu S[\Lambda] \psi = \bar{\psi} S[\Lambda]^{-1} \gamma^\mu S[\Lambda] \psi\end{aligned}$$

We claim that $S[\Lambda]^{-1} \gamma^\mu S[\Lambda] = \Lambda^\mu_\nu \gamma^\nu$

for infinitesimal version

$$(1 - \frac{1}{2} \omega_{\rho\sigma} S^{\rho\sigma}) \gamma^\mu (1 + \frac{1}{2} \omega_{\rho\sigma} S^{\rho\sigma}) = (1 + \frac{1}{2} \omega_{\rho\sigma} (M^{\rho\sigma})^\mu_\nu) \gamma^\nu$$

compare only $O(\omega)$

$$[\gamma^\mu, S^{\rho\sigma}] = (M^{\rho\sigma})^\mu_\nu \gamma^\nu$$

Since $S^{\rho\sigma} = \frac{1}{2} \gamma^\rho \gamma^\sigma - \frac{1}{2} \gamma^\sigma \gamma^\rho$,

$$[\gamma^\mu, S^{\rho\sigma}] = \frac{1}{2} [\gamma^\mu, \gamma^\rho \gamma^\sigma]$$

$$= \frac{1}{2} \gamma^\mu \gamma^\rho \gamma^\sigma - \frac{1}{2} \gamma^\rho \gamma^\sigma \gamma^\mu$$

$$= \frac{1}{2} \{ \gamma^\mu, \gamma^\rho \} \gamma^\sigma - \frac{1}{2} \gamma^\rho \gamma^\mu \gamma^\sigma - \frac{1}{2} \gamma^\sigma \{ \gamma^\mu, \gamma^\rho \} + \frac{1}{2} \gamma^\sigma \gamma^\mu \gamma^\rho$$

$$= \gamma^\sigma g^{\mu\rho} - \gamma^\rho g^{\mu\sigma}$$

$$= (\delta^\sigma_\nu g^{\mu\rho} - \delta^\rho_\nu g^{\mu\sigma}) \gamma^\nu$$

And from $(M^{\rho\sigma})^\mu_\nu = g^{\rho\mu} \delta^\sigma_\nu - g^{\sigma\mu} \delta^\rho_\nu$ (Lecture EPP3)

We get $[\gamma^\mu, S^{\rho\sigma}] = (M^{\rho\sigma})^\mu_\nu \gamma^\nu$ which prove the statement

- The vector & scalar formed by spinor representation are enough to construct the Dirac Lagrangian :

$$\mathcal{L} = \bar{\psi} (i \gamma^\mu \partial_\mu - m) \psi$$

Quantum Electrodynamics

(internal symmetry)

From the Dirac Lagrangian, there is a phase transformation which preserves $\mathcal{L}$.

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$$

$$\psi \rightarrow \psi' = e^{i\theta}\psi \quad \text{where } \theta \text{ is a constant, } \psi'^\dagger = \psi^\dagger e^{-i\theta}$$

$$\begin{aligned}\mathcal{L} \rightarrow \mathcal{L}' &= \bar{\psi}'(i\gamma^\mu \partial_\mu - m)\psi' \\ &= \bar{\psi} e^{-i\theta}(i\gamma^\mu \partial_\mu - m)e^{i\theta}\psi = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = \mathcal{L}\end{aligned}$$

- We call this $U(1)$ symmetry (Unitary with 1 dimension)

- In the case of $\theta = \text{a constant}$ = local $U(1)$ symmetry

★ Now what happens if θ is a function of space time?

Follow the previous calculation we can clearly see that

$$\begin{aligned}\psi \rightarrow e^{i\theta(x)}\psi &\Rightarrow \mathcal{L} \rightarrow \mathcal{L}' = \bar{\psi}'(i\gamma^\mu \partial_\mu - m)\psi' \\ &= \bar{\psi} e^{-i\theta(x)}(i\gamma^\mu \partial_\mu - m)e^{i\theta(x)}\psi \\ &= \bar{\psi} e^{-i\theta(x)}e^{i\theta(x)}(i\gamma^\mu \partial_\mu - m)\psi \\ &\quad + \bar{\psi} e^{-i\theta(x)}i\gamma^\mu (\partial_\mu e^{i\theta(x)})\psi \\ &= \mathcal{L} - \bar{\psi}\gamma^\mu \psi \partial_\mu \theta(x)\end{aligned}$$

This is not invariant! The term looks like $J^\mu \partial_\mu \theta(x)$ where $J^\mu = \bar{\psi}\gamma^\mu \psi$

J^μ is also the Noether's current for the global $U(1)$ symmetry $\Rightarrow \partial_\mu J^\mu = 0$

★ A fix is to add an extra field A_μ into the Lagrangian

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi + A_\mu J^\mu$$

$$\text{Let } \psi \rightarrow e^{i\theta(x)}\psi \quad \text{and } A_\mu \rightarrow A_\mu + \delta A_\mu$$

$$\mathcal{L} \rightarrow \mathcal{L}' = \mathcal{L} - \bar{\psi} \gamma^\mu \psi \partial_\mu \theta(x) + \bar{\psi} \gamma^\mu \psi \delta A_\mu$$

$\therefore$ If we want $\mathcal{L} \rightarrow \mathcal{L}$, we need $\delta A_\mu = \partial_\mu \theta$

★ The Lagrangian $\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi + eA_\mu j^\mu$

has the $U(1)$ gauge symmetry which its transformation is

$$\psi \rightarrow e^{i\theta(x)}\psi, \quad A_\mu \rightarrow A_\mu + \frac{1}{e}\partial_\mu \theta(x)$$

★ The new added field (which helps us fixing the Lagrangian) is called the gauge field.

- Surprisingly, this is the photon field. This means the full Lagrangian takes the

$$\begin{aligned} \text{form } \mathcal{L} &= \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi + eA_\mu \bar{\psi}\gamma^\mu \psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \\ &= \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad \text{where } D_\mu \equiv \partial_\mu - ieA_\mu \end{aligned}$$

Covariant derivative

- We have no time to discuss the following topics: fermion quantisation, parity, charge conjugation, gauge field quantisation, ...

Spontaneous Symmetry Breaking

- Another important concept in particle physics is breaking symmetries
- Gauge symmetry is a beautiful idea but it does not work on weak interactions the same way it works on QED.
- To give mass to gauge bosons for weak interactions, gauge symmetry must be broken (is realised in a different way)

Discrete symmetry example

- take $\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi^2)$ which has a Z_2 symmetry $\phi \rightarrow -\phi$

- Suppose the potential is $V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2$, with $\phi = \pm v$ for minima



for $V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2 = \frac{\lambda}{4}\phi^4 - \frac{\lambda v^2}{2}\phi^2 + \frac{\lambda v^4}{4} \Rightarrow m_\phi^2 = \lambda v^2$

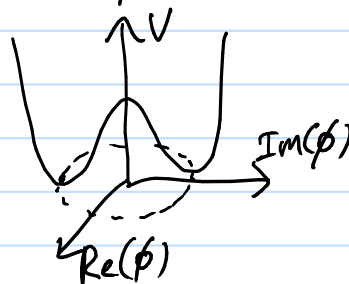
- The vacuum expectation value (vev) of this field will be the lowest energy configuration $\langle 0 | \phi | 0 \rangle \equiv \langle \phi \rangle = \pm v$
- We can define a new field η , $\phi = v + \eta$, such that $\langle \eta \rangle = 0$
- The Lagrangian becomes

$$\mathcal{L} = \frac{1}{2} \partial_\mu \eta \partial^\mu \eta - \frac{\lambda}{4} (\eta^2 + 2\eta v)^2 = \frac{1}{2} \partial_\mu \eta \partial^\mu \eta - \frac{\lambda}{4} (\eta^4 + 4\eta^2 v^2 + 4\eta^3 v)$$

- The scalar η has a mass $m_\eta^2 = 2\lambda v^2$ and no longer has a Z_2 symmetry (the cubic η^3 term)

U(1) symmetry

- Consider $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - V(\phi^* \phi)$ which is invariant under $\phi \rightarrow e^{i\theta} \phi$
- Suppose $V(\phi^* \phi) = \frac{\lambda}{4}(\phi^* \phi - v^2)^2$
- The minima form a circle $\langle \phi^* \phi \rangle = v^2 \Rightarrow \langle \phi \rangle = e^{i\alpha} v$
- The Mexican hat



- All vacua are equivalent, we can choose $\alpha = 0$ without losing generality
 $\Rightarrow \phi = \phi_1 + i\phi_2 \Rightarrow \langle \phi_1 \rangle = v \text{ \& } \langle \phi_2 \rangle = 0$
- Similar to the discrete case, we can shift $\phi_1 = v + \eta_1$, $\phi_2 = \eta_2$

- We get $V = \frac{\lambda}{4} (\eta_1^2 + 2\eta_1 v + \eta_2^2)^2$

$\therefore \mathcal{L} = \frac{1}{2} \partial_\mu \eta_1 \partial^\mu \eta_1 + \frac{1}{2} \partial_\mu \eta_2 \partial^\mu \eta_2 - \frac{\lambda}{4} (\eta_1^4 + \eta_2^4 + 4\eta_1^2 v^2 + 2\eta_1^2 \eta_2^2 + 4\eta_1^3 v)$

- $U(1)$ symmetry disappears and we now have 2 fields with masses

$$m_{\eta_1} = 2\lambda v^2 \quad \text{and} \quad m_{\eta_2} = 0$$

- The massless η_2 is called Nambu-Goldstone boson

The Higgs mechanism

- Suppose we have theory with a scalar field transformed under

$U(1)$ gauge $\phi \rightarrow e^{i\theta(x)} \phi$

- The Lagrangian is then $\mathcal{L} = (D_\mu \phi)^* (D^\mu \phi) - V(\phi^* \phi)$

where $D_\mu \equiv \partial_\mu - igA_\mu$

- The kinetic terms for $\phi = \phi_1 + i\phi_2$ become

$$\begin{aligned} (D_\mu \phi)^* (D^\mu \phi) &= [(\partial_\mu + igA_\mu)(\phi_1 - i\phi_2)] [(\partial^\mu - igA^\mu)(\phi_1 + i\phi_2)] \\ &= [\partial_\mu \phi_1 - i\partial_\mu \phi_2 + igA_\mu \phi_1 + gA_\mu \phi_2] \\ &\quad \times [\partial^\mu \phi_1 + i\partial^\mu \phi_2 - igA^\mu \phi_1 + gA^\mu \phi_2] \\ &= \partial_\mu \phi_1 \partial^\mu \phi_1 + \partial_\mu \phi_2 \partial^\mu \phi_2 + g^2 \phi_1^2 A_\mu A^\mu + g^2 \phi_2^2 A_\mu A^\mu \\ &\quad + 2gA_\mu \phi_2 \partial^\mu \phi_1 - 2gA_\mu \phi_1 \partial^\mu \phi_2 \end{aligned}$$

Now if we choose a symmetry breaking potential $V(\phi^* \phi) = \frac{\lambda}{4} (\phi^* \phi - v^2)^2$

we have $\langle \phi \rangle = v e^{i\alpha}$, $\Rightarrow$ choosing the phase to zero. The shifted

fields are $\eta_1 = \phi_1 - v$, $\eta_2 = \phi_2$

The kinetic terms become

$$(D_\mu \phi)^\dagger (D^\mu \phi) = \partial_\mu \eta_1 \partial^\mu \eta_1 + \partial_\mu \eta_2 \partial^\mu \eta_2 + g^2 \eta_1^2 A_\mu A^\mu + 2g^2 v \eta_1 A_\mu A^\mu + \underline{g^2 v^2 A_\mu A^\mu} + g^2 \eta_2^2 A_\mu A^\mu \\ + 2g A_\mu \eta_2 \partial^\mu \eta_1 - 2g A_\mu \eta_1 \partial^\mu \eta_2 - 2g A_\mu v \partial^\mu \eta_2$$

• Notice that the term $g^2 v^2 A_\mu A^\mu$ is a mass term for the gauge boson $\Rightarrow M_A = g v$

• Similar to the global case the field η_1 has a mass $m_\eta = 2\lambda v^2$

• We can write ϕ in polar form $\phi(x) = (\eta(x) + v) e^{iG(x)/v}$

$\Rightarrow$ We can use $U(1)$ transformation $\phi \rightarrow e^{-iG(x)/v} \phi$ to get rid off $G(x)$

Then the Full Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + (\partial_\mu + ig A_\mu) \phi^\dagger (\partial^\mu - ig A^\mu) \phi - \frac{\lambda}{4} (\phi^\dagger \phi - v^2)^2 \\ = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \partial_\mu \eta \partial^\mu \eta + \underline{g^2 v^2 A_\mu A^\mu} + 2g^2 v \eta A_\mu A^\mu + g^2 \eta \eta A_\mu A^\mu \\ - \frac{\lambda}{4} (\eta^4 + \underline{4\eta^2 v^2} + 2v\eta^3) \\ \text{mass} \quad \text{mass}$$

$\Rightarrow$ We start with 1 complex scalar + 1 massless gauge field

$\Rightarrow$ Now we have 1 real scalar + 1 massive gauge field

- Another real field corresponding to the Nambu-Goldstone boson has disappeared

This is called the Higgs mechanism

- We can say that the gauge boson has eaten the scalar G and grown heavy.

- G is sometimes called a "Would-be-Goldstone boson"

- Note that this way the $U(1)$ symmetry is not broken, it is spontaneously broken / hidden

which guarantees renormalisability.

Higgs mechanism on fermions

- Higgs field can also give mass to fermions
- Consider a term in a potential so-called Yukawa interaction

$$\mathcal{L} = y \bar{\psi} \psi \phi$$

- If we have a spontaneous symmetry breaking potential $\phi = \eta + \theta$

$$\mathcal{L} = y \eta \bar{\psi} \psi + \underbrace{y \theta \bar{\psi} \psi}_{\text{mass for the fermion}}$$

SU(2) symmetry

- Consider a two-component field

$$\Phi = \begin{pmatrix} \phi_a \\ \phi_b \end{pmatrix}, \quad \phi_a = \phi_1 + i\phi_2, \quad \phi_b = \phi_3 + i\phi_4$$

- We require the Lagrangian to be invariant under SU(2)

$$\Phi \rightarrow \Phi' = U \Phi, \quad U \text{ is an element of } SU(2) \text{ (} 2 \times 2 \text{ } \mathbb{C} \text{ matrix)}$$

- The Lagrangian is similar to previous case

$$\mathcal{L} = (D_\mu \Phi)^\dagger D^\mu \Phi - \frac{\Lambda}{4} (\Phi^\dagger \Phi - v^2)^2$$

- What does the local SU(2) look like?

Gauge fields for SU(2): W & Z bosons

As we have shown for Lorentz transf., any element of SU(2) can be written as an exponential function of generators (Pauli matrices) $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \dots$

$$U = e^{i\theta_i \sigma_i}, \quad \theta_i \text{ are three real parameters}$$

- Again, we can see that promoting SU(2) to local makes $\partial_\mu \Phi \partial^\mu \Phi$ not invariant
- We need gauge fields $\Rightarrow$ introduce $W_\mu^{(x)} = W_\mu^i(x) \sigma_i$

Exercise If we define $D_\mu \Phi = (\partial_\mu - ig W_\mu) \Phi$, check that

$W_\mu(x)$ transforms as $W_\mu(x) \rightarrow W'_\mu(x) = U(x) W_\mu(x) U^\dagger(x) \pm \frac{i}{g} (\partial_\mu U(x)) U^\dagger(x)$

(You need to check that $\mathcal{L}$ is invariant, also use $U^\dagger U = 1$)

- Now we are ready to break $SU(2)$

- To connect with the real world Φ is called Higgs doublet

- Symmetries we are interested in are $U(1) \times SU(2)$

- As in $V(\Phi^\dagger \Phi) = \frac{\Lambda}{4} (\Phi^\dagger \Phi - v^2)^2$, the minimum occurs at

$$\Phi^\dagger \Phi = v^2$$

- We can use gauge transformation to set $\phi_a = 0$ and $\phi_b = v$

$$\langle \Phi \rangle = \begin{pmatrix} 0 \\ v \end{pmatrix} \Rightarrow \Phi = \begin{pmatrix} 0 \\ h + v \end{pmatrix}$$

- Higgs transforms under $U(1)$ & $SU(2)$

$$D_\mu \Phi = (\partial_\mu + \frac{ig_1}{2} B_\mu + \frac{ig_2}{2} W_\mu) \Phi$$

B_μ is the gauge field of $U(1)$ (we called $U(1)_Y$ - hypercharge)

W_μ is the gauge fields of $SU(2)$ ($SU(2)_L$ - left)

g_1 & g_2 are corresponding couplings

Matrix-wise, we can rewrite W_μ as

$$\begin{aligned} W_\mu &= W_{1\mu} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + W_{2\mu} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + W_{3\mu} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} W_{3\mu} & W_{1\mu} - iW_{2\mu} \\ W_{1\mu} + iW_{2\mu} & -W_{3\mu} \end{pmatrix} \equiv \begin{pmatrix} W_{3\mu} & \sqrt{2} W_\mu^+ \\ \sqrt{2} W_\mu^- & -W_{3\mu} \end{pmatrix} \end{aligned}$$

Expanding around vacuum value gives

$$D_\mu \Phi = \begin{pmatrix} 0 \\ \partial_\mu h \end{pmatrix} + \frac{ig_1}{2} \begin{pmatrix} 0 \\ B_\mu(h+v) \end{pmatrix} + \frac{ig_2}{2} \begin{pmatrix} \sqrt{2} W_\mu^+(h+v) \\ -W_\mu^3(h+v) \end{pmatrix}$$

$$\therefore (D^\mu \Phi)^\dagger (D_\mu \Phi) = \frac{1}{2} \partial_\mu h \partial^\mu h + \frac{g_2^2}{2} W_\mu^- W^{\mu+} (h+v)^2 + \left[\frac{g_2^2}{2} W_\mu^3 W^{3\mu} - \frac{g_1 g_2}{2} W_\mu^3 B^\mu + \frac{g_1^2}{4} B_\mu B^\mu \right] (h+v)^2$$

We write $Z_\mu = \frac{g_2 W_\mu^3 - g_1 B_\mu}{\sqrt{g_1^2 + g_2^2}} \equiv W_\mu^3 \cos \theta_w - B_\mu \sin \theta_w$, $\theta_w = \text{Weinberg angle}$

$$(D^\mu \Phi)^\dagger (D_\mu \Phi) = \frac{1}{2} \partial_\mu h \partial^\mu h + \frac{g_2^2}{2} W_\mu^+ W^{\mu-} (h+v)^2 + \frac{1}{4} (g_1^2 + g_2^2) Z_\mu Z^\mu (h+v)^2$$

- Identifying $\frac{M_W^2}{2} W_\mu^+ W^{\mu-}$ & $\frac{M_Z^2}{2} Z_\mu Z^\mu$ with terms above gives

$$M_{W^\pm} = g_2 v, \quad M_Z = \frac{1}{2} \sqrt{g_1^2 + g_2^2} v$$

- This is the Higgs mechanism in the standard model

- We start with 3 + 1 gauge bosons ($SU(2) \times U(1)$) and 1 complex scalar field

- After SSB, 2 out of 3 gauge bosons ($SU(2)$) get masses & 1 gauge boson ($U(1)$) get a mass, $\Rightarrow$ With some degree of mixing, 3 gauge bosons get masses (W^+, W^-, Z)

- The massless gauge boson left from this process is photon. This is identified with the orthogonal combination (orthogonal with Z)

$$A_\mu = \frac{g_1 W_\mu^3 + g_2 B_\mu}{\sqrt{g_1^2 + g_2^2}}$$

Orthogonality can be checked by simply

taking $A_\mu Z^\mu = 0$, where $W_\mu^3 W^{3\mu} = B_\mu B^\mu = 1$ and $W_\mu^3 B^\mu = 0$

- And there is no mass term for A_μ